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Smart Traffic Signal Control in Urban Environments Leveraging SUMO and Curiosity-Enhanced MAPPO

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ABSTRACT

In urban multi-intersection traffic signal control, learning effective policies is challenging because signal decisions have delayed effects, intersections are strongly coupled and pedestrian service requirements may conflict with vehicle throughput. While multi-agent reinforcement learning provides a promising framework for coordinated signal control, most existing studies primarily emphasise inter-agent coordination and pay limited attention to exploration efficiency during training. To address this issue, this paper proposes a pedestrian-aware curiosity-enhanced multi-agent proximal policy optimisation (MAPPO) method for urban traffic signal control, in which random network distillation is incorporated as an intrinsic-reward mechanism to encourage more effective exploration. The proposed method is evaluated in SUMO on a real-map-based Helsinki traffic network with 16 signalised intersections and is compared with vanilla MAPPO and several representative traffic signal control baselines. Experimental results show that the proposed method not only accelerates training convergence but also improves key traffic performance metrics, including queue length, waiting time, throughput and pedestrian service quality. These results demonstrate the effectiveness of curiosity-driven exploration for multi-agent traffic signal control in realistic urban environments.

1 | Introduction

Urban traffic signal control remains a challenging problem in intelligent transportation systems because traffic conditions vary over time, neighbouring intersections are strongly coupled and different road users often have competing service needs. In realistic urban networks, a signal decision made at one intersection can influence downstream queues and delays only after several phase transitions and its effect may further interact with turning movements, spillback and crossing demands from pedestrians. For this reason, traffic signal control is not simply a local timing problem, but a dynamic network-level decision problem that requires adaptive coordination under uncertainty. Survey studies have shown that reinforcement learning (RL) has become an important direction for adaptive traffic signal control because it

can learn control policies directly from traffic interactions rather than relying only on manually designed timing plans or fixed coordination rules [1, 2].

The development of RL-based traffic signal control has progressed from isolated-intersection settings toward more structured and large-scale network control. Representative studies such as IntelliLight demonstrated that deep RL can effectively adjust traffic phases based on real-time traffic states, while FRAP further showed that explicitly modeling phase competition can improve learning and exploit the structural symmetry of intersections [3, 4]. These studies helped establish that learning-based controllers can outperform static or manually tuned strategies in many simulated environments, especially when traffic demand changes over time and fixed timing plans become less suitable. At the same

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time, they also revealed that the design of state representation, reward signals and action space is central to successful learning in traffic signal control.

As research attention shifted from isolated intersections to urban networks, multi-agent reinforcement learning (MARL) became increasingly important for handling non-stationarity and coordination among intersections. PressLight introduced a max-pressure-inspired learning framework to improve arterial coordination and showed the value of pressure-based reward design for network traffic efficiency [5]. CoLight then modelled inter-intersection communication through graph attention, which made it possible for each agent to adapt its control decision according to neighbouring conditions [6]. MPLight further emphasised scalability in large road networks through decentralised RL with parameter sharing and pressure-based design, while AttendLight proposed a more universal attention-based architecture intended to generalise across intersections with different road layouts and flow patterns [7, 8]. Together, these studies show that the field has moved beyond the basic question of whether learning can be applied to traffic signal control, and has increasingly focused on coordination, scalability, model structure, and transferability in realistic network settings.

Despite these advances, an important issue has received much less direct attention in traffic signal control, namely, exploration efficiency during training. Many existing RL- or MARL-based traffic signal control methods are mainly trained with extrinsic traffic rewards derived from queue length, waiting time, delay, or pressure. While such reward signals are natural and task-relevant, they do not always provide strong guidance for exploration in complex urban environments. In a multi-intersection setting, the impact of a phase decision may be delayed, spatially coupled and only partially observable from a local view. A locally reasonable action can later create congestion propagation downstream, whereas a seemingly suboptimal short-term action may improve network flow after several transitions. Under these conditions, relying only on extrinsic rewards may lead to inefficient exploration, slow convergence and suboptimal coordination policies. Although this issue is well recognised more broadly in reinforcement learning, it has not been studied as systematically in traffic signal control as questions of reward engineering or inter-agent communication [9].

A related point is that much of the existing traffic signal control literature remains predominantly vehicle-oriented. In many studies, the main optimisation targets are vehicle delay, queue length, travel time, or throughput, while pedestrian service is either simplified or omitted. However, real urban intersections must serve both vehicles and pedestrians and the corresponding control objectives are not always aligned. A phase allocation policy that is favourable for vehicle throughput may lengthen pedestrian waiting time, while frequent pedestrian service may reduce vehicle efficiency during peak demand. Pedestrian-aware studies have started to address this imbalance. For example, intelligent vehicle pedestrian light (IVPL) considered both vehicle and pedestrian flows and aimed to reduce total user delay rather than vehicle delay alone, while PV-TSC investigated joint control of pedestrian and vehicle traffic using learning-based signal control. More work has also emphasised pedestrian safety and mixed-objective optimisation in signalised intersections [10–12]. These

studies suggest that pedestrian-aware traffic control is not a minor extension, but an important step toward more realistic and socially relevant signal control policies.

The above observations motivate the investigation of exploration enhancement to improve pedestrian-aware multi-intersection traffic signal control. We focus on this question in a cooperative MARL setting, where each signalised intersection acts as an agent and coordinated learning is required across the network. To build a stable and practical learning backbone, we adopt multi-agent proximal policy optimisation (MAPPO). MAPPO is a PPO-based cooperative MARL framework that has shown strong empirical performance across several multi-agent benchmarks with relatively simple implementation and competitive sample efficiency, making it a reasonable choice when the goal is to study an additional factor such as exploration enhancement, rather than to redesign the entire coordination architecture [13].

To improve exploration during training, we incorporate random network distillation (RND) into MAPPO. RND provides an intrinsic reward based on the prediction error of a trainable network with respect to a fixed random target network, thereby encouraging the agent to visit less familiar states [9]. This mechanism is attractive in our setting for two reasons. First, it is simple to integrate into existing RL pipelines and introduces only modest computational overhead. Second, it is conceptually orthogonal to the coordination mechanism of MARL: it does not replace centralised training or decentralised execution, but instead augments the training signal so that each agent can explore its traffic state space more effectively. In the context of multi-intersection traffic signal control, this is particularly relevant because different signal decisions may produce meaningful effects only after several interaction steps and because network-level coordination may otherwise be learned slowly when the agents repeatedly visit a narrow set of familiar traffic patterns.

In addition to exploration, this paper explicitly incorporates pedestrian information into the control state and task objective. The goal is not only to improve vehicle efficiency, but also to better reflect realistic urban operation in which pedestrian crossings are a regular and important component of signal control. Compared with vehicle-only settings, the inclusion of pedestrian demand makes the control task more representative of real-world intersections and also more challenging, since the learned policy must balance multiple interacting service objectives. By combining pedestrian-aware modelling with curiosity-driven exploration, this work aims to study whether a more informative training signal can lead to better learning dynamics and stronger network-level performance in realistic traffic environments. This problem setting is consistent with calls in the literature for traffic signal control methods that are not only efficient, but also more comprehensive in the road users and control effects they consider [1, 10, 12].

Based on the above motivation, this paper proposes a pedestrian-aware curiosity-enhanced MAPPO framework for urban traffic signal control. The proposed method is evaluated in SUMO on a real-map-based Helsinki urban road network with 16 signalised intersections. We compare the proposed method with vanilla MAPPO and several representative traffic signal control baselines to examine both learning behaviour and traffic performance.

Rather than treating curiosity as a generic add-on, the purpose of this study is to investigate whether intrinsic-reward-based exploration provides practical value in a realistic MARL-based signal control problem characterised by delayed effects, coupled intersections and mixed vehicle-pedestrian demands.

The main contributions of this paper are summarised as follows:

First, we formulate pedestrian-aware multi-intersection traffic signal control as a cooperative MARL problem in which delayed and coupled traffic effects make efficient exploration an important challenge.

Second, we develop a curiosity-enhanced MAPPO framework by integrating RND into MARL training, allowing each intersection agent to receive intrinsic rewards that encourage broader and more effective exploration while preserving the standard coordinated learning structure of MAPPO.

Third, we evaluate the proposed method on a realistic SUMO traffic network derived from a real urban map and compare it with vanilla MAPPO and representative traffic signal control baselines, showing improved convergence behaviour and competitive performance in queue length, waiting time and throughput.

2 | Related work

2.1 | Surveys and General Progress in RL-Based Traffic Signal Control

Reinforcement learning has become one of the main research directions in adaptive traffic signal control because it enables policies to be learned directly from interactions with traffic environments rather than relying only on fixed timing plans or handcrafted decision rules. Survey studies have shown that RL-based traffic signal control has made substantial progress in state representation, reward design, multi-agent coordination, evaluation protocols, and large-scale deployment-oriented simulation studies [1, 2, 14]. In particular, survey studies indicate that the field has gradually moved beyond proof-of-concept studies at isolated intersections and has become increasingly concerned with scalability, robustness, heterogeneous traffic participants and more realistic network-level evaluation.

Earlier RL-based traffic signal control studies mainly focused on single-intersection settings, where the learning problem is easier to formulate and the effect of signal actions is more directly observable. Representative methods such as IntelliLight and FRAP showed that deep reinforcement learning can outperform several conventional baselines in simulated environments and that suitable structural inductive biases, such as phase competition modelling, can improve learning efficiency and signal performance. More work also began to explore richer traffic state representations and temporal modelling. For example, Ma et al. proposed a deep RL method with temporal traffic pattern mining, indicating that time-varying traffic structure can be explicitly incorporated into signal control learning [15]. These studies collectively established RL as a viable framework for adaptive traffic signal control, while also revealing that state representation and delayed traffic effects are central issues in policy learning.

Although these studies laid the foundation for RL-based traffic signal control, they do not fully address the challenges of realistic urban networks, where traffic conditions are spatially coupled, signal decisions propagate across intersections and partial observability becomes more severe. As a result, later work increasingly shifted toward multi-intersection and network-level learning, where coordination and training stability play a more central role than in isolated-intersection settings.

2.2 | Network-Level and Multi-Agent Traffic Signal Control

To address the coupling among neighbouring intersections, many studies formulate traffic signal control as an MARL problem. In this setting, each signalised intersection is treated as an agent that makes local decisions while jointly affecting network-level traffic states. A major line of work focuses on improving coordination through more suitable reward formulations, scalable training schemes and explicit communication between intersections. Representative methods such as PressLight, CoLight, MPLight and AttendLight have shown that network-level performance can be improved by combining transportation-inspired rewards, graph-based message passing, parameter sharing and attention-based modelling. These methods have become strong baselines in the RL-TSC literature and are frequently used to benchmark new approaches.

Beyond these widely cited methods, later research further explored alternative coordination structures. Xu et al. proposed a hierarchical and cooperative learning framework for traffic signal control, showing that hierarchical decomposition can help manage both local efficiency and broader network objectives in multi-intersection settings [16]. This line of work suggests that the challenge in network-level traffic signal control is not merely whether intersections should coordinate, but also how coordination should be structured to balance local responsiveness and global efficiency. Related studies have likewise examined decentralised control, cooperative value decomposition and architecture design choices for large-scale traffic networks.

In addition to coordination, reliability under realistic operating conditions has also received increasing attention. For example, robustness-oriented studies have investigated how RL-based traffic signal control models behave under perturbed structures or less idealised settings. Xu et al. proposed a structure-compression-based method to improve the robustness of deep RL traffic signal control, reflecting a broader trend in the field toward models that are not only effective under nominal simulation settings but also more stable and reliable under practical uncertainty [17]. This trend is important because it reflects a broader shift in the literature toward training behavior and generalization, rather than focusing only on average reward in a single benchmark scenario.

However, despite substantial progress in coordination, communication and scalability, most network-level RL-TSC studies still focus mainly on state design, reward engineering, architecture choice, or transferability. Comparatively less attention has been paid to the question of whether policy learning itself is limited by inefficient exploration in complex and coupled traffic

environments. This gap becomes particularly relevant when delayed action effects and local observations make external traffic rewards less informative during training.

2.3 | Pedestrian-Aware Traffic Signal Control

A large portion of the RL-based traffic signal control literature remains predominantly vehicle-centric. In many studies, the main optimisation targets are vehicle delay, queue length, throughput, or pressure, whereas pedestrian demand is simplified, weakly modelled, or omitted altogether. This simplification is convenient for algorithm design, but it limits the realism of the learned control policy because actual urban intersections must allocate service to both vehicles and pedestrians and the corresponding objectives are not always aligned. Survey studies have also noted that more comprehensive traffic control formulations should consider heterogeneous road users rather than focusing only on vehicle flow [2, 14].

Several studies have started to address this limitation. Yazdani et al. proposed the IVPL, a deep RL-based adaptive signal control framework that jointly considers vehicles and pedestrians and aims to minimise total user delay rather than vehicle delay alone [10]. Their results show that pedestrian-aware control can change the preferred signal allocation policy, especially in settings with irregular pedestrian demand or jaywalking behaviour. Xu et al. proposed PV-TSC, which explicitly models both pedestrian and vehicle traffic in a distributed reinforcement learning framework and extends pressure-style reasoning to mixed traffic participants [11]. These works indicate that pedestrian-aware signal control is not a minor variation of conventional RL-TSC, but a more realistic reformulation of the control objective.

Subsequent studies have further expanded this direction. Ren et al. studied a two-step deep reinforcement learning approach that aims to improve both pedestrian safety and overall traffic efficiency, showing that pedestrian-aware signal control can be framed not only in terms of delay reduction but also in terms of safety-sensitive decision making [12]. Poudel et al. proposed a joint pedestrian and vehicle traffic optimisation framework for urban environments, again emphasising that signal control performance should be evaluated under mixed traffic objectives rather than vehicle-only criteria [18]. Compared with the extensive literature on vehicle-oriented RL-TSC, however, pedestrian-aware studies remain relatively limited, especially in large network-level multi-agent settings.

This gap is directly relevant to the present work. Since pedestrian crossing demand influences phase allocation and may conflict with vehicle progression, pedestrian-aware optimisation makes the traffic signal control task both more realistic and more challenging. It also provides a more suitable setting for testing whether better training signals can improve policy learning in complex urban control scenarios.

2.4 | Exploration and Learning Efficiency in RL/MARL

Efficient exploration is a long-standing challenge in reinforcement learning, especially in environments where the conse-

quences of actions are delayed, sparse, or only partially reflected in immediate rewards. Among exploration methods based on intrinsic motivation, RND has become a widely used approach because it provides a simple prediction-error-based novelty bonus and can be integrated into standard RL pipelines with relatively low additional complexity. In general RL tasks, RND has been shown to improve exploration by encouraging the agent to visit less familiar states, particularly when external reward signals alone are insufficient to guide learning effectively [19].

In the multi-agent setting, the exploration problem can be even more difficult because the state-action space grows rapidly and the learning dynamics of one agent affect the observations and rewards seen by others. Efficient exploration remains a central challenge in MARL, and a body of studies has explicitly investigated exploration efficiency and training stability in cooperative multi-agent environments. This broader MARL literature is relevant to traffic signal control because urban multi-intersection control also involves delayed consequences, strong interdependence among agents and repeated visitation of familiar traffic regimes during training.

By contrast, in the traffic signal control literature, exploration is usually discussed only indirectly through reward shaping, state design, or learning curves. Most RL-TSC methods optimise policies using external traffic objectives such as queue length, waiting time, or pressure, but do not directly study whether the agents are exploring the traffic state space efficiently during training. This omission may be important in large and coupled traffic networks, where local observations can be incomplete and the network-level effect of a signal decision may become visible only after several downstream transitions. From this perspective, intrinsic-reward-based exploration appears to be a relevant yet still under-studied direction for traffic signal control, especially when the control objective also includes heterogeneous road users such as pedestrians [2, 14].

Motivated by these observations, our work focuses on the intersection of two relatively underexplored directions in the traffic signal control literature: pedestrian-aware multi-agent signal optimisation and intrinsic-reward-based exploration enhancement. Rather than proposing a new coordination architecture from scratch, we study whether adding curiosity-driven exploration to a stable cooperative MARL backbone can improve training efficiency and traffic performance in a realistic urban network setting.

3 | Methodology

3.1 | Problem Formulation

Urban multi-intersection traffic signal control is formulated in this paper as a cooperative MARL problem. Consider a road network with N signalised intersections. Each intersection is modelled as one control agent and all agents interact with the same traffic environment. Since each intersection only has access to its local traffic information during decision making, while the overall traffic evolution depends on the joint actions of all intersections, the task is naturally modelled as a decentralised partially observable Markov decision process.

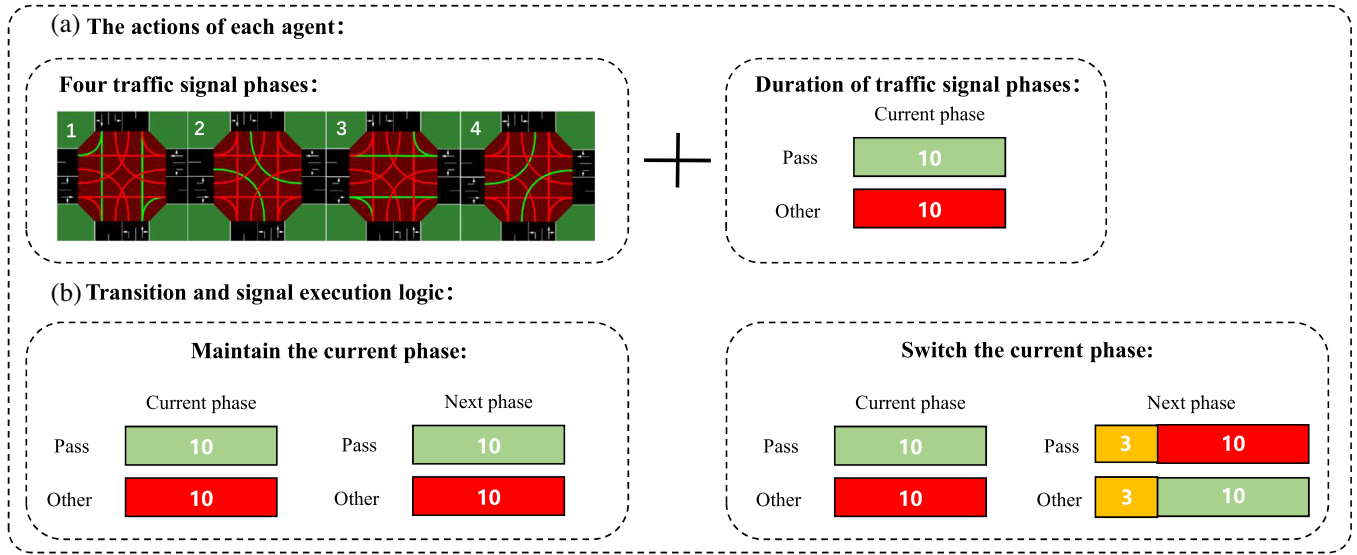


FIGURE 1 | Action design and phase transition logic.

$$\mathcal{G} = \langle I, S, \{O_i\}_{i=1}^N, \{A_i\}_{i=1}^N, P, R, \gamma \rangle$$

where $I = \{1, \dots, N\}$ denotes the set of intersection agents, S is the global traffic state, O_i and A_i are the local observation and action spaces of agent i , $P(s_{t+1}|s_t, a_t)$ is the environment transition function, r_t is the shared training reward and $\gamma \in (0, 1)$ is the discount factor. This formulation is consistent with the cooperative multi-agent setting commonly considered in MAPPO-style learning, where local policies are executed in a decentralised way while coordination is learned through centralised training.

At each decision step t , each agent i receives a local observation $o_i^t \in O_i$ and selects an action $a_i^t \in A_i$. The joint action $\mathbf{a}_t = (a_1^t, \dots, a_N^t)$ is then applied to the traffic environment, which produces the next state s_{t+1} and the reward r_t . The goal is to learn a joint policy π that maximises the expected discounted cumulative reward over the control horizon:

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]$$

Because all intersections contribute to the same network-level control objective, the learning problem is cooperative rather than competitive. In the later algorithm design, each agent executes its policy using only local observations, whereas centralised information is used only during training to improve value estimation and coordination, following the standard centralised-training and decentralised-execution paradigm in cooperative MARL.

3.2 | Traffic Environment Modelling

3.2.1 | Agent and Phase Definition

Each signalised intersection in the target road network is modelled as one control agent. This formulation is consistent with network-level traffic signal control, where each intersection makes local decisions while jointly affecting the traffic evolution

of the entire network. At each decision step, the agent i selects one signal phase from a predefined set of feasible green phases:

$$a_i^t \in A_i = \{1, 2, \dots, M\}$$

where M denotes the number of admissible phases at the intersection i and a_i^t is the phase selected at the time step t . The joint network action is denoted by $\mathbf{a}_t = (a_1^t, \dots, a_N^t)$. The feasible phase situation at the intersection for this study is shown in Figure 1.

The action space is restricted to feasible signal phases rather than arbitrary light combinations, so that the learned control policy remains consistent with practical traffic signal operation. Safety-related transition rules, such as yellow intervals between conflicting green phases, are handled by the simulator instead of being directly optimised by the agent. This design follows the common practice in RL-based traffic signal control, where the learning problem is simplified to phase-level decision making while preserving basic operational realism. Survey studies also note that action abstraction is an important component of practical RL-TSC system design [1, 14].

3.2.2 | Observation Design

The local observation of the agent i is constructed to describe both traffic demand and signal control context:

$$o_i^t = [q_i^t, w_i^t, p_i^t, \phi_i^t, g_i^t]$$

where q_i^t denotes vehicle queue length, w_i^t denotes vehicle queue time, p_i^t represents pedestrian queue time, ϕ_i^t is the current signal phase and g_i^t denotes the current phase duration.

As shown in Figure 2, this observation design is motivated by the structure of the traffic signal control task. Queue and waiting variables reflect local congestion and service pressure, which are directly relevant to signal timing decisions. Pedestrian demand

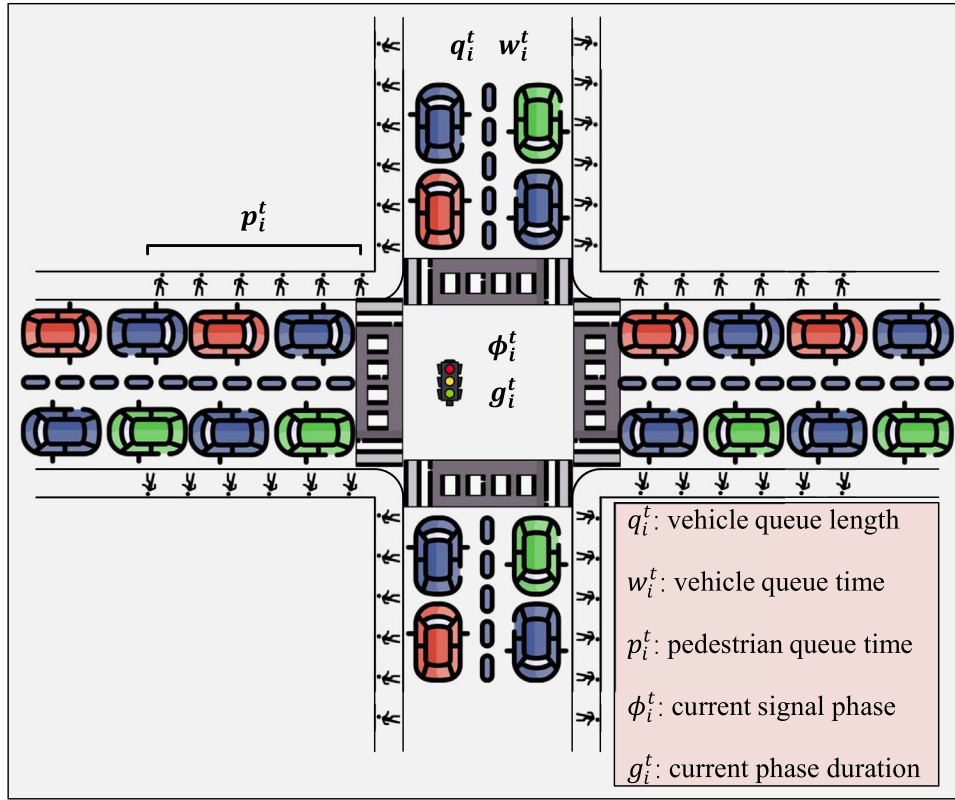


FIGURE 2 | Intersection-level observation.

is explicitly included because this work considers pedestrian-aware control rather than a purely vehicle-oriented setting. In addition, the current phase and elapsed phase time provide temporal context for phase switching and help the agent avoid unrealistic decisions that ignore the recent signal history. Existing reviews of RL-based traffic signal control emphasise that state representation is one of the most influential factors in policy quality, since it determines how well the agent can infer both current congestion and short-term control context [1].

The inclusion of pedestrian information is also supported by recent pedestrian-aware traffic signal control studies. These studies show that signal policies optimised only for vehicle traffic may fail to allocate green time appropriately when pedestrian crossing demand is substantial, whereas explicitly incorporating pedestrian-related observations can produce more balanced control behaviour under mixed traffic conditions [10].

3.2.3 | Transition and Signal Execution Logic

After the joint action $\mathbf{a}_t$ is selected, the corresponding phase commands are executed in the SUMO environment. Let ϕ_i^t denote the currently active phase at the intersection i . If the selected phase a_i^t is the same as the current phase, the simulator keeps the current green phase active. If $a_i^t \neq \phi_i^t$, the simulator first applies the predefined transition logic, including the yellow interval required for safe phase switching and then activates the selected green phase.

This execution mechanism makes the learned policy compatible with practical signal control constraints while keeping the action space compact. It also prevents the policy from directly producing physically unsafe phase combinations. From a modelling perspective, the agent learns phase-level control decisions, whereas low-level transition safety is enforced by the simulator. This separation is useful because it reduces the complexity of the learning problem without removing the main traffic control challenge.

3.2.4 | Extrinsic Reward Design

To jointly evaluate vehicle efficiency and pedestrian service, we define a traffic cost function at the time step t as

$$C_t = \alpha Q_t^{\text{veh}} + \beta W_t^{\text{veh}} + \eta Q_t^{\text{ped}} + \zeta W_t^{\text{ped}}$$

where Q_t^{veh} and W_t^{veh} denote vehicle queue length and vehicle waiting time, respectively and Q_t^{ped} and W_t^{ped} denote pedestrian queue length and pedestrian waiting time. The weighting coefficients α , β , η and ζ are used to balance the contributions of vehicle- and pedestrian-related terms.

Based on this cost, the extrinsic reward is defined in a dense temporal-difference form:

$$r_t^{\text{ext}} = C_t - C_{t+1}$$

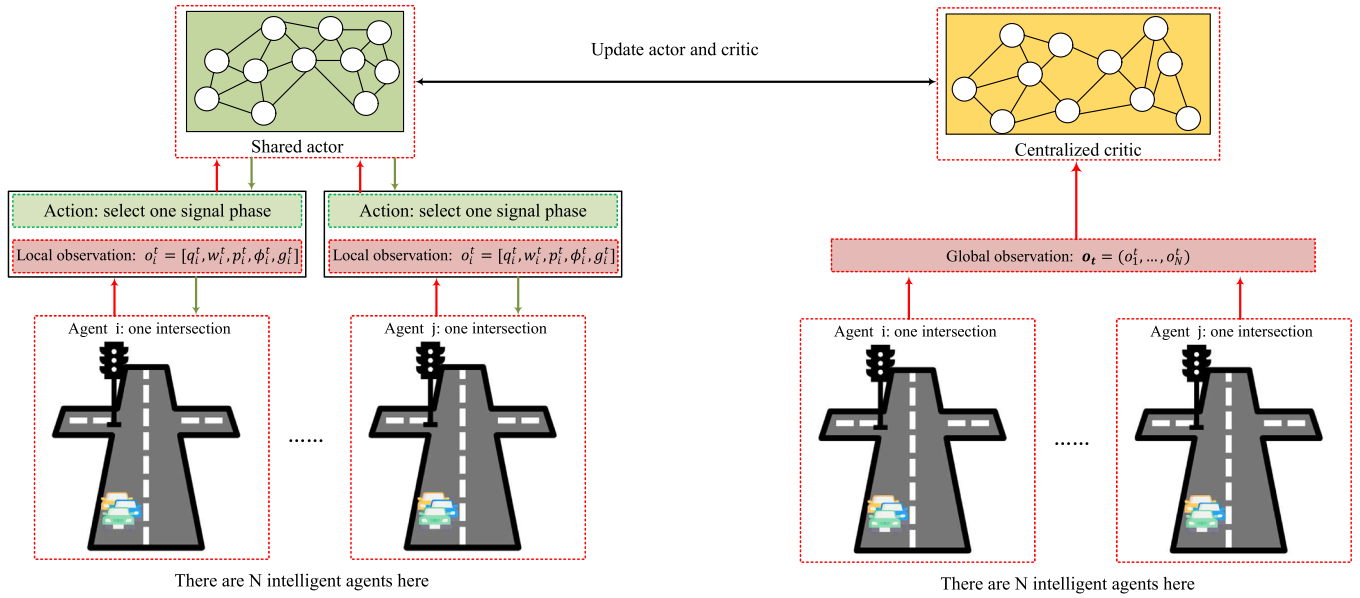


FIGURE 3 | CTDE structure for traffic signal control.

Under this definition, the reward is positive when the overall traffic cost decreases and negative when congestion or waiting becomes worse. Compared with a reward that considers only vehicle delay or queue length, this formulation is more consistent with pedestrian-aware traffic signal control because it treats pedestrian service as part of the control objective rather than as an auxiliary constraint. Pedestrian-aware studies likewise emphasise that mixed vehicle-pedestrian optimisation can lead to signal policies that are qualitatively different from purely vehicle-oriented control [12].

The reward is designed to provide relatively dense learning feedback while still reflecting the network-level control goal. This is important in traffic signal control, where delayed action effects can make sparse or overly coarse reward definitions difficult to optimise effectively.

3.3 | MAPPO for Cooperative Traffic Signal Control

As shown in Figure 3, this work adopts MAPPO as the cooperative learning backbone to learn coordinated control policies across intersections. The choice of MAPPO is motivated by the structure of the traffic signal control problem considered here. At execution time, each intersection only has access to its local traffic observation, so the policy must operate in a decentralised manner. During training, however, value estimation can benefit from broader information across the network because the quality of one agent's decision depends on the behaviour of neighbouring intersections. MAPPO follows the centralised-training and decentralised-execution paradigm and has been shown to perform strongly in cooperative multi-agent learning while retaining the stability and simplicity of PPO-style optimisation.

For each intersection agent i , the decentralised actor selects an action according to its local observation o_i^t :

$$a_i^t \sim \pi_\theta(a_i^t | o_i^t)$$

where θ denotes the shared actor parameters. During training, a centralised critic is used to estimate the value of the joint traffic condition $V_\phi(\mathbf{o}_t)$, where $\mathbf{o}_t = (o_1^t, \dots, o_N^t)$ is the collection of local observations from all agents and ϕ denotes the critical parameters. In this way, policy execution remains decentralised, while value learning is supported by more informative network-level context. This design is well-suited to multi-intersection traffic signal control, where local control decisions are coupled through shared traffic dynamics.

To reduce the variance of policy gradient estimation, generalised advantage estimation (GAE) is used. The temporal-difference residual is defined as $\delta_t = r_t + \gamma V_\phi(\mathbf{o}_{t+1}) - V_\phi(\mathbf{o}_t)$ and the corresponding advantage estimate is computed by $\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma \lambda_{\text{gae}})^l \delta_{t+l}$, where λ_{gae} is the GAE parameter. GAE is used to obtain a practical trade-off between bias and variance, which is helpful for stabilising policy-gradient learning in environments with delayed and noisy reward signals [20].

The actor is updated using the PPO clipped surrogate objective. Let $\rho_t(\theta) = \frac{\pi_\theta(a_t | o_t)}{\pi_{\theta_{\text{old}}}(a_t | o_t)}$ denote the probability ratio between the current policy and the behaviour policy used to collect the rollout data. The policy objective is then written as

$$L^{\text{clip}}(\theta) = \mathbb{E}_t [\min(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t)]$$

where ϵ is the clipping coefficient. This objective prevents excessively large policy updates and is one of the main reasons PPO achieves stable optimisation in practice. The value function is trained by minimising the regression loss

$$L^{\text{value}}(\phi) = \mathbb{E}_t [(V_\phi(\mathbf{o}_t) - \hat{R}_t)^2]$$

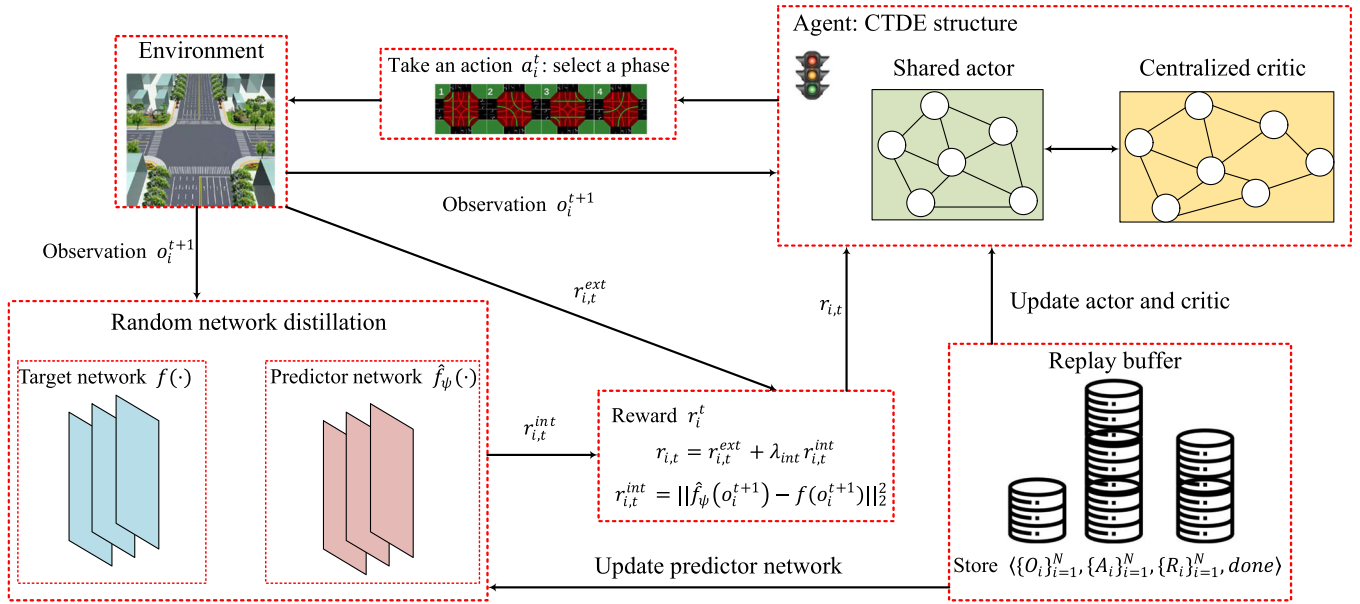


FIGURE 4 | RND-enhanced learning pipeline for one agent.

where $\hat{R}_t$ is the return target. Following standard actor-critic training, the overall optimisation objective is

$$L = -L^{\text{clip}} + c_v L^{\text{value}} - c_e H(\pi_\theta)$$

where $H(\pi_\theta)$ is the entropy regularisation term and c_v and c_e are the weighting coefficients for the value loss and entropy term, respectively

In the proposed framework, MAPPO provides a stable cooperative optimisation backbone for pedestrian-aware multi-intersection traffic signal control. However, standard MAPPO still relies on the informativeness of the task reward during training. In complex urban environments, where the effect of a signal decision may be delayed and coupled across intersections, extrinsic traffic rewards alone may not always guide exploration efficiently. This observation motivates the curiosity-driven exploration enhancement introduced in the next subsection.

3.4 | Curiosity-Enhanced Exploration With Random Network Distillation

Although MAPPO provides a stable cooperative learning backbone, policy improvement in standard MAPPO still depends on the informativeness of the task reward during training. In pedestrian-aware multi-intersection traffic signal control, this can become a limitation. The effect of a signal phase decision is often delayed rather than immediately visible, since traffic released at one intersection may only influence downstream congestion after several future transitions. In addition, each agent observes only its local traffic condition, while the overall network state evolves through coupled interactions among neighbouring intersections. When pedestrian service is also included in the control objective, the learning problem becomes more complex because the agent must balance multiple interacting traffic demands. Under these conditions, the extrinsic traffic reward may be insufficient to

guide efficient exploration in the early stage of training, even when the cooperative learning framework itself is well designed. Intrinsic-reward-based exploration has been shown to be effective when external feedback alone is not sufficiently informative [9].

To improve exploration efficiency, we incorporate RND into the MAPPO training process, as illustrated in Figure 4. RND introduces a novelty-based intrinsic reward using two networks: a fixed, randomly initialised target network $f(\cdot)$ and a trainable predictor network $\hat{f}_\psi(\cdot)$. Given the next local observation o_i^{t+1} of agent i , the intrinsic reward is defined as

$$r_{i,t}^{\text{int}} = \|\hat{f}_\psi(o_i^{t+1}) - f(o_i^{t+1})\|_2^2$$

where a larger prediction error indicates that the current traffic observation is less familiar to the predictor network and is therefore treated as more novel. In this way, the agent is encouraged to explore traffic states that have not been frequently visited during training. RND is particularly attractive in this setting because it is simple to implement, introduces only limited additional computation and can be combined directly with standard reinforcement learning objectives.

The predictor network is trained to match the output of the fixed target network by minimising

$$L^{\text{RND}}(\psi) = \mathbb{E}_o \left[\|\hat{f}_\psi(o) - f(o)\|_2^2 \right]$$

As training proceeds, observations that are repeatedly encountered become easier for the predictor to approximate and their intrinsic reward gradually decreases. To combine task-oriented learning with exploration enhancement, the total reward used for policy optimisation is defined as

$$r_{i,t} = r_{i,t}^{\text{ext}} + \lambda_{\text{int}} r_{i,t}^{\text{int}}$$

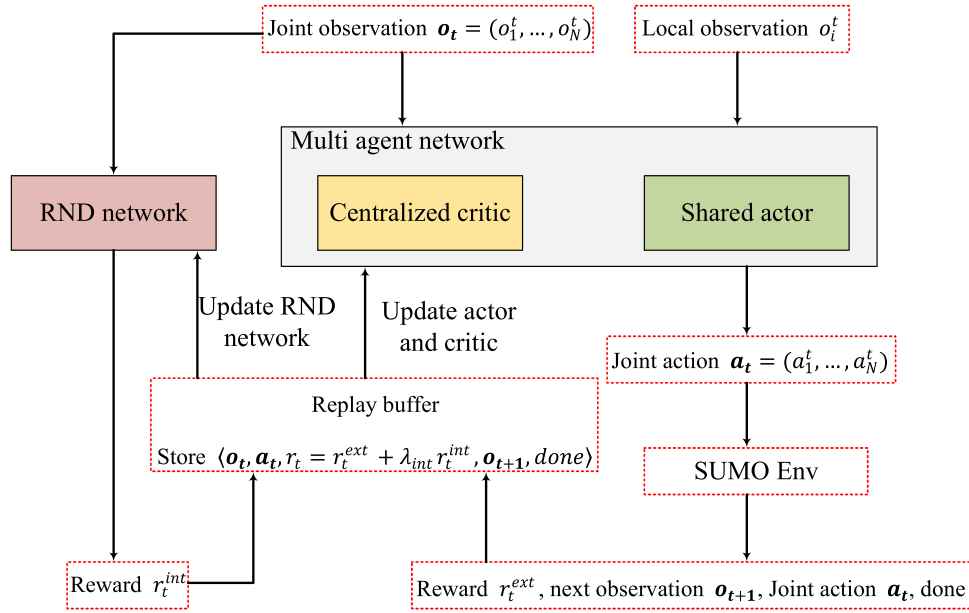


FIGURE 5 | Overall training pipeline of RND-MAPPO.

where $r_{i,t}^{\text{ext}}$ is the extrinsic traffic reward defined in Section 3.2, $r_{i,t}^{\text{int}}$ is the intrinsic reward produced by RND and λ_{int} controls the contribution of the curiosity term. Under this formulation, the extrinsic reward keeps the policy aligned with the traffic control objective, while the intrinsic reward provides an auxiliary signal that encourages broader exploration during training. This design follows the standard use of RND, in which intrinsic and extrinsic rewards are combined rather than optimised separately.

In the proposed framework, RND is applied independently to the local observation stream of each intersection agent instead of to a centralised global state. This design is consistent with decentralised execution, keeps the exploration signal local and scalable and avoids introducing an additional centralised exploration module on top of MAPPO. It also makes the method easier to integrate into network-level traffic signal control, where each agent naturally interacts with its own local traffic pattern. During deployment, the learned traffic signal policy operates without any curiosity bonus and the final controller depends only on observed traffic conditions. Therefore, RND is used only as a training-time exploration aid and does not change the execution form of the learned policy.

3.5 | Training Procedure and Implementation Details

As shown in Figure 5, the proposed RND-MAPPO framework is trained through repeated interaction with the SUMO traffic environment. At each decision step, every intersection agent receives its local observation and selects a phase action using the decentralised actor policy. The joint action is then executed in the simulator, which produces the next traffic state and the extrinsic traffic reward defined in Section 3.2. In parallel, the RND module computes an intrinsic reward for each agent based on the prediction error on the next local observation. The extrinsic and intrinsic rewards are then combined to form the reward signal

used for training. In this way, the policy is guided not only by traffic-efficiency feedback but also by an auxiliary novelty signal that encourages broader exploration during learning. RND was originally proposed precisely as a lightweight way to augment standard reinforcement learning with intrinsic rewards and this property makes it suitable for integration with PPO-style policy optimisation.

After one rollout segment is collected, the centralised critic evaluates the collected trajectories and the return target is computed as

$$\hat{R}_t = \sum_{l=0}^{T-t-1} \gamma^l r_{t+l}$$

where r_t denotes the reward used for policy optimisation. In the proposed method, this reward includes the curiosity-enhanced signal defined in Section 3.4. Based on the critic estimates and the collected rewards, GAE is applied to obtain the advantage targets and MAPPO updates are then performed for both the actor and the critic. The actor is optimised with the PPO clipped objective, while the critic is updated by value regression. This training procedure follows the standard centralised-training and decentralised-execution setting of MAPPO, which has been shown to provide stable cooperative learning in multi-agent environments.

At the same time, the RND predictor network is updated using the observations collected from the rollout, whereas the target network remains fixed throughout training. Therefore, the full optimisation loop contains two coupled update processes: a MAPPO update that improves the traffic signal control policy and a predictor update that gradually reduces the novelty error on frequently visited observations. As a result, traffic states that have already been explored repeatedly contribute less intrinsic reward over time, while less familiar traffic states continue to encourage exploration. This mechanism allows curiosity to

ALGORITHM 1 | Training Procedure of the Proposed RND-MAPPO Framework.

Require: Traffic environment E ; actor parameters θ ; critic parameters ϕ ; fixed RND target network f_{tar} ; RND predictor parameters φ ; intrinsic reward coefficient λ_{int}

Ensure: Trained decentralised actor policy π_θ

- 1: Initialise actor policy π_θ
- 2: Initialize critic network V_ϕ
- 3: Initialise RND predictor network f_φ
- 4: Fix the parameters of the RND target network f_{tar}
- 5: for each training iteration do
- 6: Reset environment E
- 7: Obtain initial local observations $\mathbf{o}_t = (o_1^t, \dots, o_N^t)$
- 8: for $t = 0$ to $T - 1$ do
- 9: for each agent $i = 1$ to N do
- 10: Sample action $a_i^t \sim \pi_\theta(\cdot | o_i^t)$
- 11: end for
- 12: Execute the joint action $\mathbf{a}_t = (a_1^t, \dots, a_N^t)$ in E
- 13: Observe next observations $\mathbf{o}_{t+1}$ and extrinsic reward r_t^{ext}
- 14: Compute intrinsic reward r_t^{int} using the RND prediction error
- 15: Compute total reward:

$$r_t = r_t^{ext} + \lambda_{int} r_t^{int}$$
- 16: Store transition $(\mathbf{o}_t, \mathbf{a}_t, r_t, \mathbf{o}_{t+1}, done)$
- 17: end for
- 18: Compute value estimates using V_ϕ
- 19: Compute return targets
- 20: Compute advantage estimates
- 21: Update actor parameters θ using the MAPPO objective
- 22: Update critic parameters ϕ using the value loss
- 23: Update RND predictor parameters φ using collected observations
- 24: end for
- 25: return π_θ

influence training without changing the execution form of the learned traffic controller. Based on the above design, the complete training procedure of the proposed RND-MAPPO framework is summarized in Algorithm 1.

During deployment, only the decentralised actor policy is used to select traffic phases. The centralised critic and the RND module are needed only during training and are not required at test time. Hence, the final controller remains a standard phase-selection policy conditioned on observed traffic states, while curiosity serves only as a training-time aid for improving exploration and learning efficiency.

4 | Experimental Setup

4.1 | Study Area and Simulation Platform

All experiments are conducted in the Eclipse SUMO microscopic traffic simulator. SUMO is an open-source, continuous and inter-modal traffic simulation platform that supports the modelling of large road networks and multiple traffic participants, including both vehicles and pedestrians. These properties make it suitable for evaluating adaptive traffic signal control in realistic urban settings, especially when the control objective involves mixed vehicle-pedestrian traffic rather than vehicle-only flow.

To evaluate the proposed method in a realistic city environment, we use a Helsinki traffic simulation context derived from the Helsinki city-area SUMO modelling framework reported in [21], as shown in Figure 6. This simulation context provides a realistic basis for studying urban traffic signal control beyond simplified synthetic benchmarks.

From this broader Helsinki simulation framework, we select a subnetwork containing 16 signalised intersections as the main study area. This choice reflects a balance between realism and experimental tractability. Compared with isolated single-intersection settings, the selected subnetwork contains nontrivial inter-intersection coupling and supports the study of coordinated multi-agent control. At the same time, it remains computationally manageable for repeated training, ablation and multi-seed evaluation. Since the selected area also includes pedestrian crossings, it is well-suited to the pedestrian-aware traffic signal control setting considered in this paper.

4.2 | Traffic Network and Signal Control Configuration

The selected Helsinki subnetwork contains 16 signalised intersections and each intersection is controlled as an individual decision unit. Although each agent makes a local signal control decision, the intersections are not independent. Vehicle flows propagate through upstream and downstream links and congestion at one junction can influence queue formation and discharge behaviour at neighbouring intersections. Therefore, the network provides a suitable testbed for coordinated multi-agent traffic signal control rather than isolated single-intersection optimisation.

For traffic signal operation, each intersection is modelled with a four-phase control plan. The four admissible green phases correspond to: (1) east-west through movements, (2) east-west protected left-turn movements, (3) north-south through movements and (4) north-south protected left-turn movements. This phase design captures the main conflicting traffic movements while keeping the control space compact enough for reinforcement learning. Right-turn movements are treated as permissive and remain effectively available under all signal phases, so that the control policy focuses on the main phase allocation problem instead of modelling detailed right-turn protection logic. Such phase-level action abstraction is common in RL-based traffic signal control because it provides a practical balance between operational realism and learning tractability.

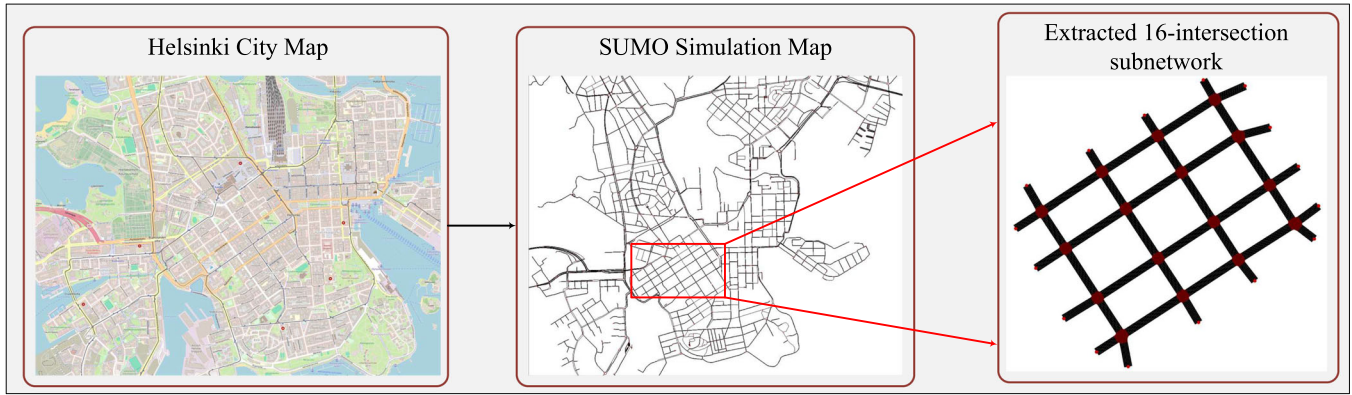


FIGURE 6 | Study area and simulation network.

To ensure realistic signal execution, phase transitions follow predefined timing rules in the simulator. When the selected phase differs from the currently active phase, the simulator inserts a 3-second yellow interval before activating the next green phase. In addition, a minimum green duration is imposed to avoid unrealistic rapid switching and to preserve stable traffic discharge within each phase. The signal controller, therefore, learns high-level phase selection decisions, while low-level transition safety is enforced by the simulator. This configuration is important because it prevents physically unsafe switching behavior and makes the learned policy more consistent with practical urban traffic signal operation.

4.3 | Traffic Demand and Pedestrian Flow Generation

The traffic demand in the selected Helsinki subnetwork is generated within the same simulation context described in Section 4.1. In this work, the selected subnetwork is populated with time-varying traffic demand consistent with the Helsinki urban simulation setting, so that the evaluation is conducted in a realistic city-scale traffic context rather than on a purely synthetic toy example.

To evaluate the proposed method under different congestion conditions, we define three separate demand settings, namely low, medium and high demand. Each simulation episode lasts 3600 s and within each episode, the traffic demand is time-varying rather than constant. Let $D_{\text{base}}^{\text{veh}}(t)$ denote the base vehicle demand profile over time. The demand level m is then defined as

$$D_m^{\text{veh}}(t) = \kappa_m D_{\text{base}}^{\text{veh}}(t), m \in \{\text{low, medium, high}\}$$

where κ_m is the scaling factor for the corresponding traffic level. In this study, as shown in Table 1, the scaling factors are set to $\kappa_{\text{low}} = 0.5$, $\kappa_{\text{medium}} = 1.0$, $\kappa_{\text{high}} = 2.0$. This design preserves the temporal and spatial structure of the base demand while allowing the experiments to cover uncongested, moderately loaded and more congested traffic regimes. Within each demand level, microscopic vehicle departures are instantiated stochastically from the time-varying arrival profile so that different simulation runs correspond to different realisations of the same nominal traffic condition.

TABLE 1 | Traffic demand scenario settings.

Scenario	Vehicle demand scaling factor
Low	0.5
Medium	1.0
High	2.0

Pedestrian demand is generated separately for the signalised crossings associated with the controlled intersections. Since SUMO supports intermodal traffic simulation, including person trips and pedestrian-related demand definitions, pedestrian arrivals can be incorporated into the same simulation environment as vehicle flows. In our setting, pedestrian demand is specified through crossing-rate profiles and is sampled stochastically over time, so that the environment reflects mixed vehicle–pedestrian service requirements rather than vehicle-only operation. This is important because the objective of the present study is not only to reduce vehicle congestion, but also to evaluate signal control under pedestrian-aware traffic conditions.

To account for simulation stochasticity, different random seeds are used to generate multiple realisations of vehicle and pedestrian arrivals under the same nominal demand level. Training and evaluation are therefore not based on a single deterministic traffic instance. Instead, the compared methods are assessed across repeated demand realisations, which improves the reliability of the evaluation and reduces the risk of drawing conclusions from one particular traffic pattern. This is consistent with discussions in the RL-based traffic control literature, which emphasise that careful evaluation protocols are necessary because the stochastic nature of both microscopic simulation and RL training can strongly affect observed performance [22].

4.4 | Compared Methods

To evaluate the proposed method in a meaningful and well-structured way, we compare it with representative network-level traffic signal control methods from several major design directions in the RL-based TSC literature. The compared methods are selected to cover different ways of improving multi-intersection control, including pressure-based coordination, graph-based

TABLE 2 | Summary of compared methods.

Method	Main idea	Role in comparison
RND-MAPPO	MAPPO + RND intrinsic reward	Proposed method
MAPPO	Cooperative PPO-based MARL	Direct backbone baseline
PressLight	Max-pressure-inspired coordination	Pressure-based baseline
CoLight	Graph-attention cooperation	Communication-based baseline
MPLight	Decentralised scalable RL	Scalability baseline
AttendLight	Attention-based transferable modelling	Attention-based baseline

communication, scalable decentralised learning and attention-based signal modelling. In addition, vanilla MAPPO is included as a direct backbone baseline so that the effect of curiosity-enhanced exploration can be examined without changing the underlying cooperative MARL framework. Survey studies on RL-based traffic signal control identify these families as representative directions in network-level signal control research and evaluation [1, 2].

The proposed method, RND-MAPPO, is compared with the following baselines. MAPPO [13] is used as the most direct comparison because it shares the same cooperative actor-critic backbone as the proposed method and differs only in the absence of the curiosity-driven exploration module. PressLight [5] is included as a representative pressure-based traffic signal control method, where the coordination mechanism is closely related to max-pressure principles. CoLight [6] is included as a representative communication-based RL method that models inter-intersection cooperation through graph attention. MPLight [7] is adopted as a representative scalable decentralised RL method for large traffic networks. AttendLight [8] is included as an attention-based baseline representing transferable intersection modelling. A summary of the compared methods is shown in Table 2.

To ensure fair comparison, all learning-based methods are evaluated on the same Helsinki subnetwork, under the same demand scenarios and with the same signal phase configuration whenever applicable. The compared methods are trained under comparable interaction budgets and hyperparameters are initialised according to the original papers or publicly available implementations when possible, followed by limited pilot tuning for stable training. This comparison protocol is intended to ensure that performance differences reflect the relative effectiveness of the methods rather than inconsistencies in the simulation environment or training budget. In particular, the comparison between MAPPO and RND-MAPPO is designed to isolate the contribution of curiosity-enhanced exploration under the same cooperative MARL backbone.

4.5 | Training Protocol and Hyperparameter Settings

All learning-based methods are trained under a unified experimental protocol as far as possible. Specifically, the compared methods use the same Helsinki 16-intersection subnetwork, the same demand scenarios and the same signal control configuration described in the previous subsections. As shown in Table 3,

TABLE 3 | Fair comparison protocol.

Item	Setting
Network	Same Helsinki 16-intersection subnetwork
Demand scenarios	Same low/medium/high settings
Action space	Same four-phase configuration
Evaluation horizon	Same for all methods
Random seeds	Multiple seeds for each learning-based method

each training episode is run over the same simulation horizon and all methods are allocated comparable environment interaction budgets so that the comparison reflects methodological differences rather than differences in training resources. In addition, all main results are obtained from multiple random seeds in order to reduce the influence of stochastic variation in both reinforcement learning and microscopic traffic simulation.

For fairness, hyperparameters are initialised according to the original papers or publicly available implementations whenever possible. They are then adjusted only through limited pilot tuning to ensure stable convergence in the selected simulation environment, rather than being aggressively optimised for a single method. This setting is particularly important in multi-agent traffic signal control, where learning performance can be sensitive to the optimiser, rollout length, entropy regularisation, and reward scale. Therefore, the goal of hyperparameter selection in this study is to provide a stable and comparable training protocol across methods, not to maximise the performance of one method through excessive tuning.

For the proposed RND-MAPPO framework, the main training hyperparameters include the actor learning rate, critic learning rate, discount factor, GAE parameter, PPO clipping coefficient, rollout length, batch size, entropy coefficient, value-loss coefficient and the intrinsic reward coefficient used for the RND module. The full hyperparameter settings are summarised in Table 4.

4.6 | Evaluation Metrics

To evaluate the proposed method from multiple perspectives, we use three groups of metrics: traffic-efficiency metrics,

TABLE 4 | Main training hyperparameters.

Parameter	Value
Episode length	3600
Total training episodes	500
Number of seeds K	5
Actor learning rate	1e-5
Critic learning rate	5e-4
Discount factor γ	0.99
GAE parameter λ_{gae}	0.95
PPO clip coefficient ϵ	0.2
Rollout length	3600
Batch size	256
Intrinsic reward coefficient λ_{int}	0.2

pedestrian-related metrics and learning-related metrics [23]. This multi-view evaluation is necessary because the goal of the present study is not only to improve final traffic performance, but also to examine whether curiosity-enhanced exploration leads to better learning efficiency in pedestrian-aware multi-intersection traffic signal control.

For traffic efficiency, we report the average queue length, the average vehicle waiting time and throughput. Let Q_t denote the total number of queued vehicles in the network at time step t . The average queue length is defined as

$$L_q = \frac{1}{T} \sum_{t=1}^T Q_t$$

where T is the evaluation horizon. A smaller value of L_q indicates lighter congestion. The average vehicle waiting time is computed as

$$T_w^{\text{veh}} = \frac{1}{M^{\text{veh}}} \sum_{m=1}^{M^{\text{veh}}} w_m^{\text{veh}}$$

where M^{veh} is the total number of evaluated vehicles and w_m^{veh} is the waiting time of the vehicle m . A smaller T_w^{veh} indicates better vehicle service quality. Throughput is defined as the total number of vehicles that successfully leave the network during the evaluation period,

$$T_p = \sum_{t=1}^T n_t^{\text{out}}$$

where n_t^{out} denotes the number of vehicles leaving the network at time step t . A larger throughput indicates better network-level traffic efficiency. These metrics are widely used in RL-based traffic signal control to evaluate congestion mitigation and traffic movement quality.

Since this study explicitly considers pedestrian-aware control, pedestrian performance is also evaluated. We report the average pedestrian waiting time and the average number of waiting pedes-

trians at controlled crossings. The average pedestrian waiting time is defined analogously to the vehicle waiting time and reflects the service quality experienced by pedestrians under a given signal control policy. The average number of waiting pedestrians measures the pedestrian service pressure at crossings during the evaluation horizon. Smaller values for both metrics indicate better pedestrian service. These indicators are included because a method that improves vehicle flow while consistently increasing pedestrian delay would not be consistent with the objective of pedestrian-aware traffic signal control. Pedestrian-aware TSC studies similarly evaluate pedestrian waiting or mixed user delay rather than relying only on vehicle-oriented metrics.

To evaluate learning efficiency, we report the training reward curve, the area under the learning curve (AUC) and the number of episodes required to reach a predefined performance threshold. The reward curve describes the learning dynamics over training episodes, while the AUC summarises overall training efficiency over a fixed horizon:

$$\text{AUC} = \sum_{e=1}^E R_e$$

where R_e is the episode reward at the training episode e and E is the total number of training episodes considered in the comparison [24].

$$\text{Episodes to Threshold} = \min \{t | R_t \geq R_{th}\}$$

$$R_{th} = 0.8 \times \max_m R_{\text{final}}^{(m)}$$

In addition, the episodes-to-threshold metric measures how quickly a method reaches a target performance level. The performance threshold is defined as 80% of the final mean reward achieved by RND-MAPPO and the number of episodes required by each method to first reach this common threshold is reported. $R_{\text{final}}^{(\text{RND-MAPPO})}$ represents the final reward of RND-MAPPO, while R_{th} represents the reference reward level. These metrics are important in the present work because the proposed method is intended not only to improve final control quality but also to enhance exploration and speed up learning. This directly addresses whether curiosity-driven exploration provides practical value beyond the final performance of the learned controller. The evaluation metrics used in the experiments are summarised in Table 5.

4.7 | Ablation and Robustness Studies

In addition to the main benchmark comparison, a set of ablation and robustness studies is designed to better understand where the performance gains come from and under what conditions the proposed method remains effective. This is important because the proposed framework combines two key ideas, namely pedestrian-aware traffic signal control and curiosity-enhanced exploration and the main comparison alone is not sufficient to isolate the contribution of each component.

The first ablation study examines the effect of curiosity-driven exploration. For this purpose, the proposed RND-MAPPO

TABLE 5 | Evaluation metrics used in the experiments.

Metric	Meaning	Better direction
Average queue length	Network congestion pressure	Lower
Average vehicle waiting time	Vehicle service quality	Lower
Throughput	Vehicles exiting the network	Higher
Average pedestrian waiting time	Pedestrian service quality	Lower
Average waiting pedestrians	Pedestrian crossing pressure	Lower
AUC of reward curve	Overall training efficiency	Higher
Episodes to threshold	Speed of convergence	Lower

TABLE 6 | Ablation and robustness study design.

Study	Compared settings	Main purpose
Effect of RND	MAPPO vs RND-MAPPO	Isolate curiosity-driven exploration
Effect of pedestrian-aware modelling	Vehicle-only vs vehicle–pedestrian vs vehicle–pedestrian + RND	Evaluate pedestrian-aware formulation
Demand robustness	Low (0.5×), medium (1.0×), high (2.0×)	Test performance across traffic regimes
Sensitivity to λ_{int}	Different intrinsic reward coefficients	Evaluate hyperparameter robustness

method is directly compared with vanilla MAPPO under the same network, demand scenarios and training protocol. Since both methods share the same cooperative actor–critic backbone, this comparison isolates the contribution of the RND exploration module. The second ablation study examines the effect of pedestrian-aware modelling. In this study, three settings are compared: a vehicle-only formulation, a vehicle–pedestrian formulation without RND and the full vehicle–pedestrian formulation with RND. This design makes it possible to evaluate whether pedestrian-aware observation and reward design improve signal control quality and whether curiosity-driven exploration remains beneficial in the more complex mixed vehicle–pedestrian setting. Pedestrian-aware traffic signal control studies suggest that mixed-objective signal control can lead to policies that differ substantially from vehicle-only optimisation, which further motivates this analysis.

To evaluate robustness, the compared methods are tested under three traffic-demand settings, namely low, medium and high demand, corresponding to the scaling factors 0.5, 1.0 and 2.0 defined in Section 4.3. This analysis is intended to examine whether the proposed method remains effective under different congestion levels rather than only in a single nominal traffic condition. In particular, it allows us to test whether curiosity-enhanced exploration becomes more useful when the traffic environment is more congested and the interaction among intersections becomes more difficult to manage.

In addition, a sensitivity study is conducted on the intrinsic reward coefficient λ_{int} used by the RND module. By comparing multiple values of λ_{int} , this experiment evaluates whether the performance of the proposed method depends strongly on a narrow hyperparameter choice or remains stable within a reasonable range. The design of the ablation and robustness studies is summarised in Table 6.

5 | Results and Discussion

5.1 | Overall Performance in the Helsinki Network

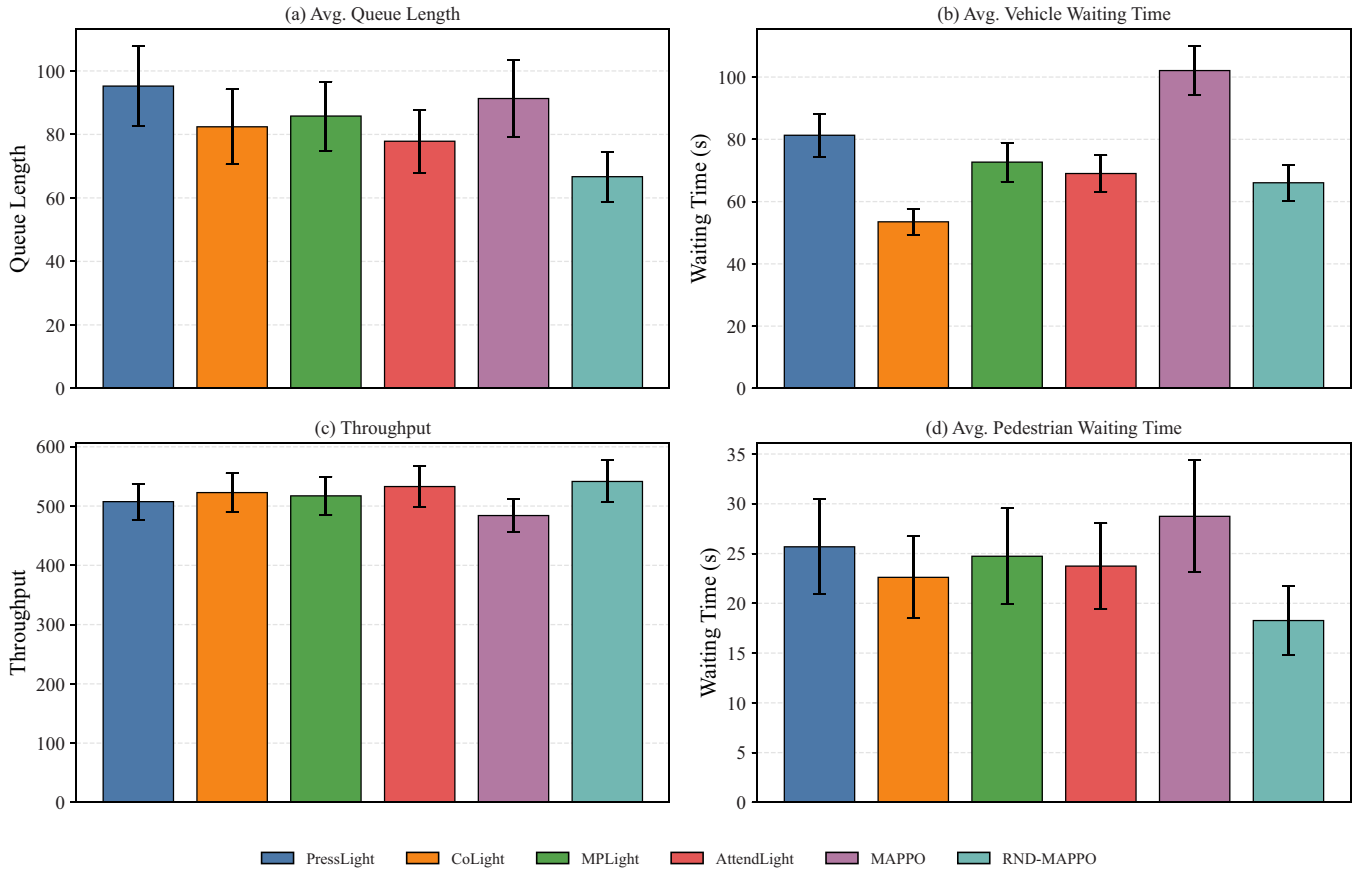
Table 7 and Figure 7 report the overall performance comparison of the proposed method and the baseline methods on the Helsinki 16-intersection network. The evaluation includes both traffic-efficiency metrics and pedestrian-related metrics, so that the comparison reflects not only vehicle flow performance but also the quality of pedestrian service under mixed traffic demand.

Overall, RND-MAPPO achieves the best performance across the main evaluation metrics. Compared with vanilla MAPPO, the proposed method reduces average queue length by 26.96%, reduces average vehicle waiting time by 35.30%, improves throughput by 11.88% and reduces average pedestrian waiting time by 36.43%. Compared with representative non-MAPPO baselines, including PressLight, CoLight, MPLight and AttendLight, the proposed method still maintains consistent advantages. In particular, the reduction in average queue length ranges from 14.34% to 29.97%, while the throughput improvement ranges from 1.61% to 6.71%. Similar gains are also observed in pedestrian-related metrics. These results suggest that the proposed framework improves not only final traffic efficiency but also the balance between vehicle and pedestrian service.

The comparison with MAPPO is particularly important because the two methods share the same cooperative MARL backbone. The observed gains, therefore, suggest that the curiosity-driven exploration module improves the quality of the learned policy rather than merely benefiting from a different coordination architecture. At the same time, the improvement in pedestrian-related metrics indicates that the proposed method does not achieve better vehicle performance at the cost of pedestrian service.

TABLE 7 | Overall performance comparison on the Helsinki 16-intersection network.

Method	Avg. queue length	Avg. vehicle waiting time	Throughput	Avg. pedestrian waiting time	Avg. waiting pedestrians
PressLight	95.23 $\pm$ 12.47	81.31 $\pm$ 6.84	507.46 $\pm$ 30.14	25.68 $\pm$ 4.77	12.58 $\pm$ 2.96
CoLight	82.41 $\pm$ 11.79	53.51 $\pm$ 4.17	522.74 $\pm$ 32.76	22.61 $\pm$ 4.12	9.41 $\pm$ 2.14
MPLight	85.79 $\pm$ 10.88	72.68 $\pm$ 6.29	517.25 $\pm$ 32.17	24.73 $\pm$ 4.82	12.13 $\pm$ 2.75
AttendLight	77.86 $\pm$ 9.87	69.02 $\pm$ 5.87	532.93 $\pm$ 34.81	23.74 $\pm$ 4.31	10.69 $\pm$ 2.48
MAPPO	91.31 $\pm$ 12.15	102.09 $\pm$ 7.73	484.01 $\pm$ 27.96	28.74 $\pm$ 5.61	14.17 $\pm$ 3.27
RND-MAPPO	66.69 $\pm$ 7.84	66.05 $\pm$ 5.71	541.51 $\pm$ 35.39	18.27 $\pm$ 3.47	7.67 $\pm$ 1.92

**FIGURE 7** | Overall comparison of traffic and pedestrian performance.

Instead, the pedestrian-aware formulation and the exploration-enhanced training process together lead to a more balanced and effective signal control strategy in the Helsinki network.

5.2 | Learning Curves and Sample Efficiency

Figure 8 shows the learning curves of the compared methods on the Helsinki 16-intersection network and Table 8 summarises the corresponding sample-efficiency indicators. While several methods improve steadily over the course of training, the proposed RND-MAPPO reaches a high-reward regime noticeably earlier than the baselines and maintains smaller fluctuations after convergence. In particular, the proposed method reaches

the predefined reward threshold after approximately 92 training episodes, whereas vanilla MAPPO requires around 150 episodes to reach a comparable level. Similar differences can also be observed relative to the strongest non-MAPPO baselines. Figure 9 zooms into the first 100 training iterations to highlight the early-stage learning behaviour. The proposed RND-MAPPO exhibits faster reward improvement during the initial training stage.

The same trend is reflected in the quantitative sample-efficiency results. Under the same training horizon, RND-MAPPO achieves the largest AUC, which is approximately 21.0% higher than that of vanilla MAPPO. At the same time, the number of episodes required to reach the predefined performance threshold is reduced from 150.17 to 92.61, corresponding to a reduction of

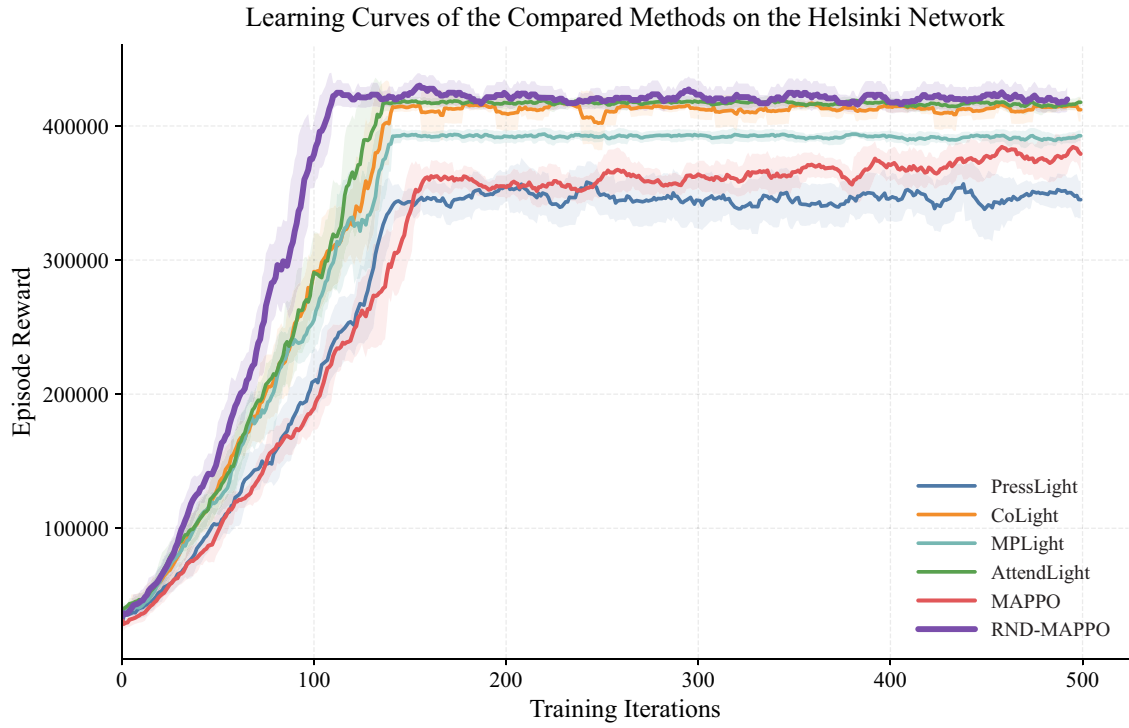


FIGURE 8 | Learning curves of the compared methods on the Helsinki network.

TABLE 8 | Sample-efficiency comparison.

Method	AUC($\times 10^8$)	Episodes to threshold	Final reward($\times 10^5$)
PressLight	1.457 ± 0.0728	138.73 ± 4.17	3.492 ± 0.175
CoLight	1.756 ± 0.0878	122.41 ± 3.79	4.142 ± 0.215
MPLight	1.670 ± 0.0919	128.37 ± 4.49	3.914 ± 0.188
AttendLight	1.781 ± 0.0926	115.86 ± 3.24	4.164 ± 0.212
MAPPO	1.505 ± 0.0737	150.17 ± 4.35	3.809 ± 0.171
RND-MAPPO	1.821 ± 0.0859	92.61 ± 3.31	4.188 ± 0.214

about 38.3%. The final reward is also improved by approximately 10.0%. Taken together, these results show that the proposed method is not only better in final traffic performance, but also substantially more sample-efficient under the same interaction budget. Here, the threshold is the common reference level defined in Section 4.6, 80% of the final mean reward of RND-MAPPO.

These findings support the main claim of the paper. The advantage of curiosity-enhanced exploration is therefore reflected not only in final reward, but also in how quickly a useful signal control policy can be learned in a complex urban environment. In simulation-based traffic signal controller development, faster convergence means that the controller reaches an effective operating regime earlier under the same training budget, which is particularly valuable in multi-intersection settings where policy learning is more difficult and computationally expensive.

A likely reason for this behaviour is that, in pedestrian-aware multi-intersection traffic signal control, the extrinsic traffic reward can be delayed and locally ambiguous. The RND module

provides an additional novelty-based learning signal in the early stage of training, encouraging the agents to explore a broader range of traffic states instead of repeatedly revisiting familiar but potentially suboptimal control patterns. As training proceeds, this broader exploration appears to translate into both faster convergence and more stable learning dynamics.

5.3 | Effect of Curiosity-Driven Exploration

To isolate the contribution of the curiosity mechanism, this subsection directly compares RND-MAPPO with vanilla MAPPO under the same cooperative MARL backbone. This comparison is intentionally designed as a controlled ablation: both methods use the same actor-critic architecture, the same traffic environment, the same reward structure and the same training budget and differ only in whether the RND-based intrinsic reward is included during training. Therefore, the observed differences in this subsection can be attributed specifically to the effect of curiosity-enhanced exploration.

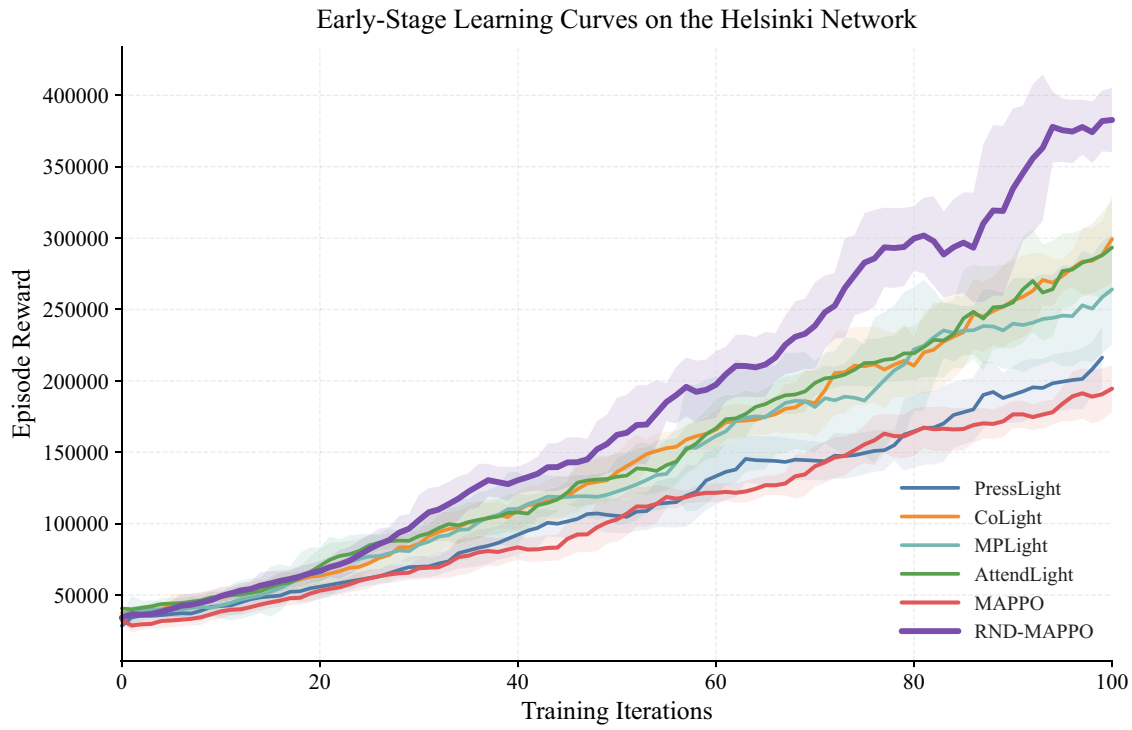


FIGURE 9 | Early-stage learning curves of the compared methods on the Helsinki network.

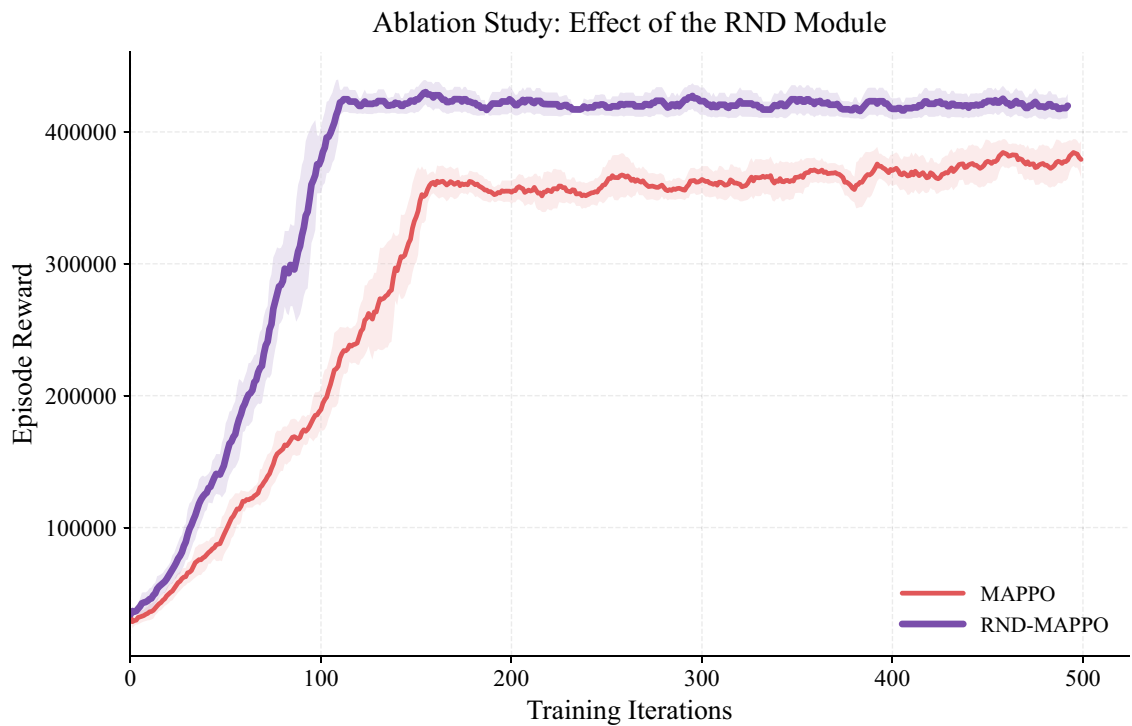


FIGURE 10 | Learning dynamics of MAPPO and RND-MAPPO.

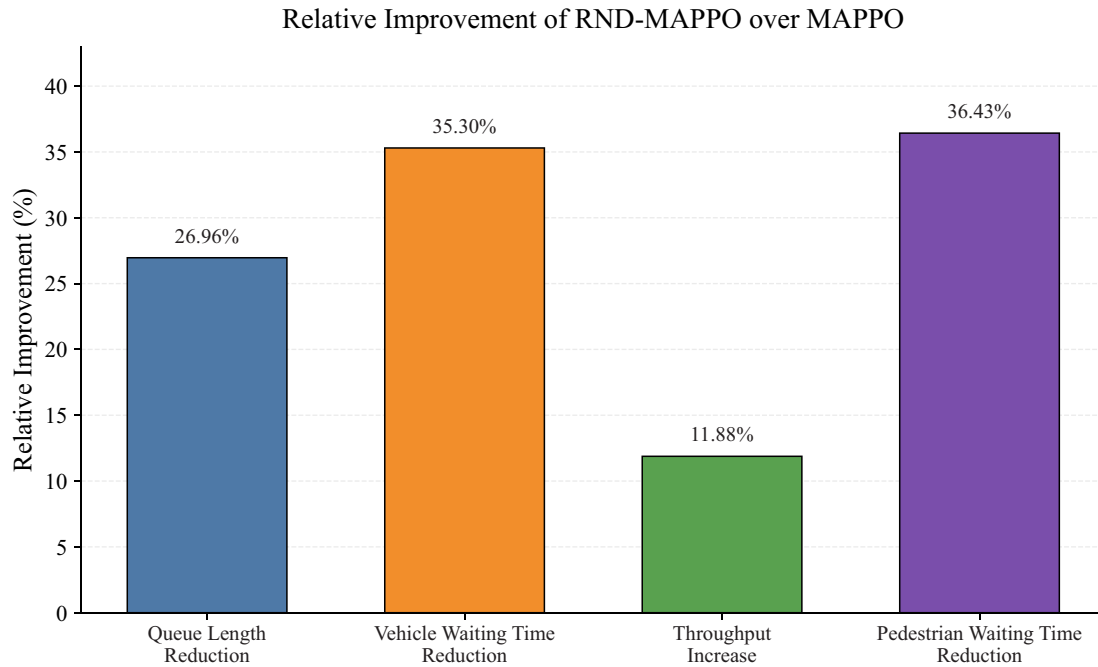
Figure 10 and Table 9 show that introducing RND changes the learning process in a consistent way. Compared with vanilla MAPPO, RND-MAPPO reaches the target performance level in fewer training episodes, achieves an AUC that is about 21.0% higher and reduces episodes-to-threshold by approximately 38.3%, as shown in Figure 11. These results indicate that the main

effect of the curiosity module is first reflected in improved sample efficiency and more stable training dynamics.

This improvement in learning dynamics also translates into a better final control policy. Under the same cooperative MARL backbone, RND-MAPPO still achieves lower queue length and waiting

TABLE 9 | Ablation study on the effect of curiosity-driven exploration.

Method	Avg. queue length	Avg. vehicle waiting time	Throughput	Avg. pedestrian waiting time	AUC($\times 10^8$)	Episodes to threshold
MAPPO	91.31 $\pm$ 12.15	102.09 $\pm$ 7.73	484.01 $\pm$ 27.96	28.74 $\pm$ 5.61	1.505 $\pm$ 0.0737	150.17 $\pm$ 4.35
RND-MAPPO	66.69 $\pm$ 7.84	66.05 $\pm$ 5.71	541.51 $\pm$ 35.39	18.27 $\pm$ 3.47	1.821 $\pm$ 0.0859	92.61 $\pm$ 3.31

**FIGURE 11** | Relative improvement of RND-MAPPO over MAPPO.**TABLE 10** | Ablation study on pedestrian-aware modelling.

Setting	Avg. queue length	Avg. vehicle waiting time	Throughput	Avg. pedestrian waiting time	Avg. waiting pedestrians
Vehicle-only MAPPO	59.14 $\pm$ 6.51	57.28 $\pm$ 5.44	586.73 $\pm$ 35.21	42.81 $\pm$ 7.72	18.63 $\pm$ 3.41
Vehicle-pedestrian MAPPO	91.31 $\pm$ 12.15	102.09 $\pm$ 7.73	484.01 $\pm$ 27.96	28.74 $\pm$ 5.61	14.17 $\pm$ 3.27
Vehicle-pedestrian RND-MAPPO	66.69 $\pm$ 7.84	66.05 $\pm$ 5.71	541.51 $\pm$ 35.39	18.27 $\pm$ 3.47	7.67 $\pm$ 1.92

time, together with higher throughput and better pedestrian service, than vanilla MAPPO. This suggests that the curiosity module does not simply accelerate early reward accumulation but helps the agents discover more effective coordination behaviours during training.

5.4 | Effect of Pedestrian-Aware Modelling

To evaluate the effect of explicitly modelling pedestrian demand, we compare three formulations: Vehicle-only MAPPO, Vehicle-Pedestrian MAPPO and the full Vehicle-Pedestrian RND-MAPPO model. This ablation is important because the contribution of pedestrian-aware modelling should be assessed not only

in terms of pedestrian service, but also in terms of the resulting trade-off with vehicle efficiency.

As shown in Table 10, incorporating pedestrian information substantially improves pedestrian service, but also changes the optimisation trade-off relative to the vehicle-only formulation. Compared with vehicle-only MAPPO, vehicle-pedestrian MAPPO markedly reduces average pedestrian waiting time and the number of waiting pedestrians, but it also increases vehicle waiting time and reduces throughput. This indicates that pedestrian-aware control is not a cosmetic extension; rather, it changes the effective objective of the learned policy and encourages more balanced phase allocation across road-user groups.

Effect of Pedestrian-Aware Modeling on Vehicle and Pedestrian Metrics

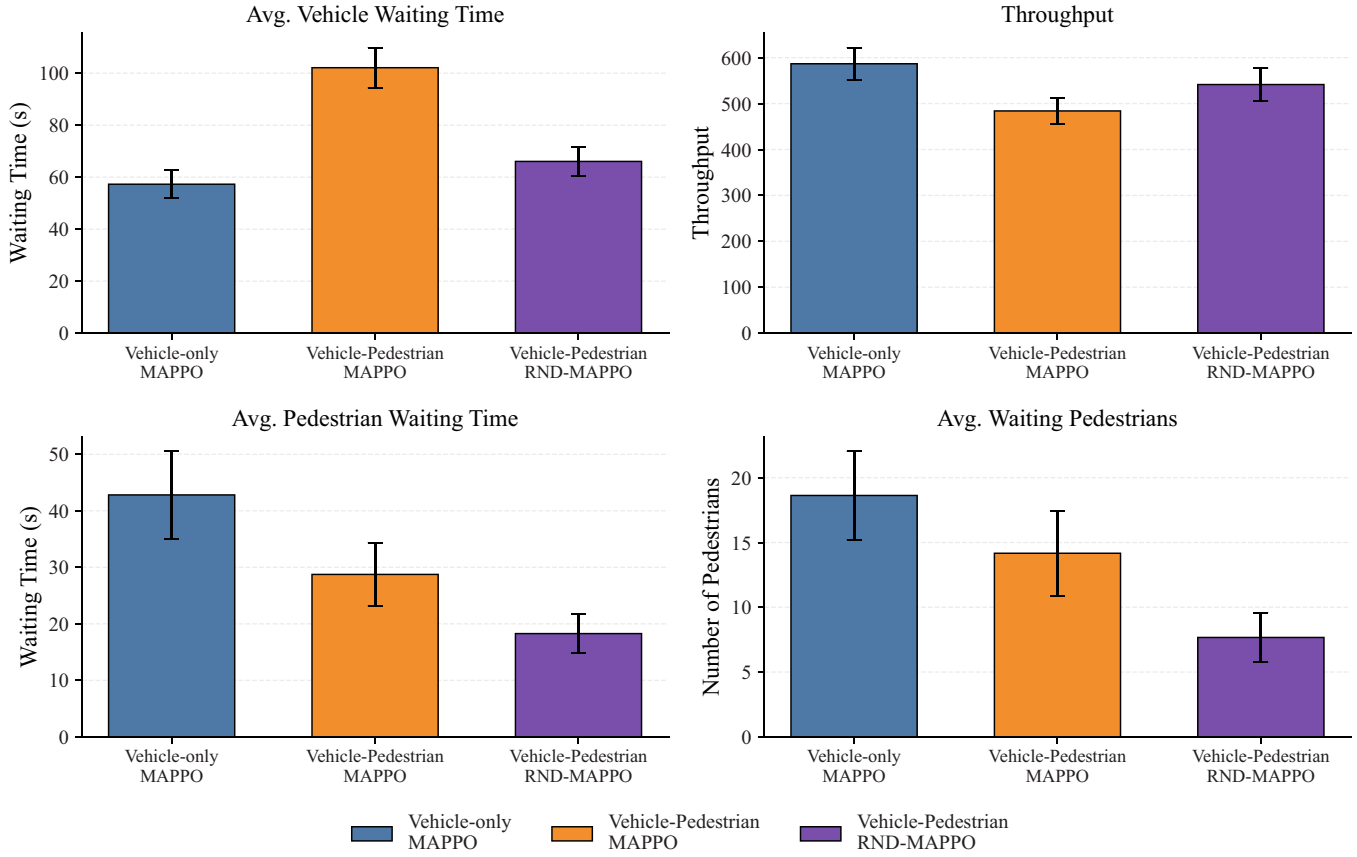


FIGURE 12 | Effect of pedestrian-aware modelling on vehicle and pedestrian metrics.

Quantitatively, moving from the vehicle-only formulation to the vehicle-pedestrian formulation reduces average pedestrian waiting time by approximately 32.9% and the average number of waiting pedestrians by about 23.9%, but also introduces a clear reduction in vehicle-oriented efficiency, as shown in Figure 12. This confirms that pedestrian-aware control changes the effective optimisation target rather than simply adding auxiliary variables.

Importantly, when RND is further introduced, the full vehicle-pedestrian RND-MAPPO model recovers a large part of the lost vehicle efficiency while continuing to improve pedestrian service, as shown in Figure 13. Therefore, the full model achieves a substantially better vehicle-pedestrian balance than either the vehicle-only formulation or the pedestrian-aware formulation without curiosity. This result supports the claim that pedestrian-aware modelling is a meaningful component of realistic multi-intersection traffic signal control.

5.5 | Robustness Under Different Demand Levels

To evaluate whether the proposed method remains effective under different traffic loads, we compare the methods under the three demand settings defined in Section 4.3, namely low demand (0.5 $\times$), medium demand (1.0 $\times$) and high demand (2.0 $\times$). This analysis is important because urban traffic conditions vary substantially over time and a method that performs well only

under one nominal demand level may not be suitable for realistic deployment. For clarity, Table 11 and Figure 14 report the demand-level comparison for vanilla MAPPO, the proposed RND-MAPPO and the strongest competing non-MAPPO baseline, namely AttendLight.

The results show that RND-MAPPO performs consistently well across all three traffic regimes. Under low demand, the proposed method already achieves the best or near-best performance, although the gap relative to the other methods is relatively modest. Under medium demand, the advantage becomes clearer across queue length, waiting time and pedestrian-related metrics. Under high demand, where congestion is most severe, the proposed method achieves the strongest improvement over the compared methods, indicating that its benefit is not limited to easier traffic conditions.

More specifically, as shown in Figure 15, compared with vanilla MAPPO, the proposed method reduces average queue length by 21.63% under low demand, 26.96% under medium demand and 35.08% under high demand. The same pattern is observed for vehicle waiting time and pedestrian service, with the largest reductions appearing under the high-demand setting. The throughput gain remains positive across all three demand levels, reaching 19.24%, 11.88% and 16.02% under low, medium and high demand, respectively. These results suggest that the proposed method remains effective across different traffic loads and becomes

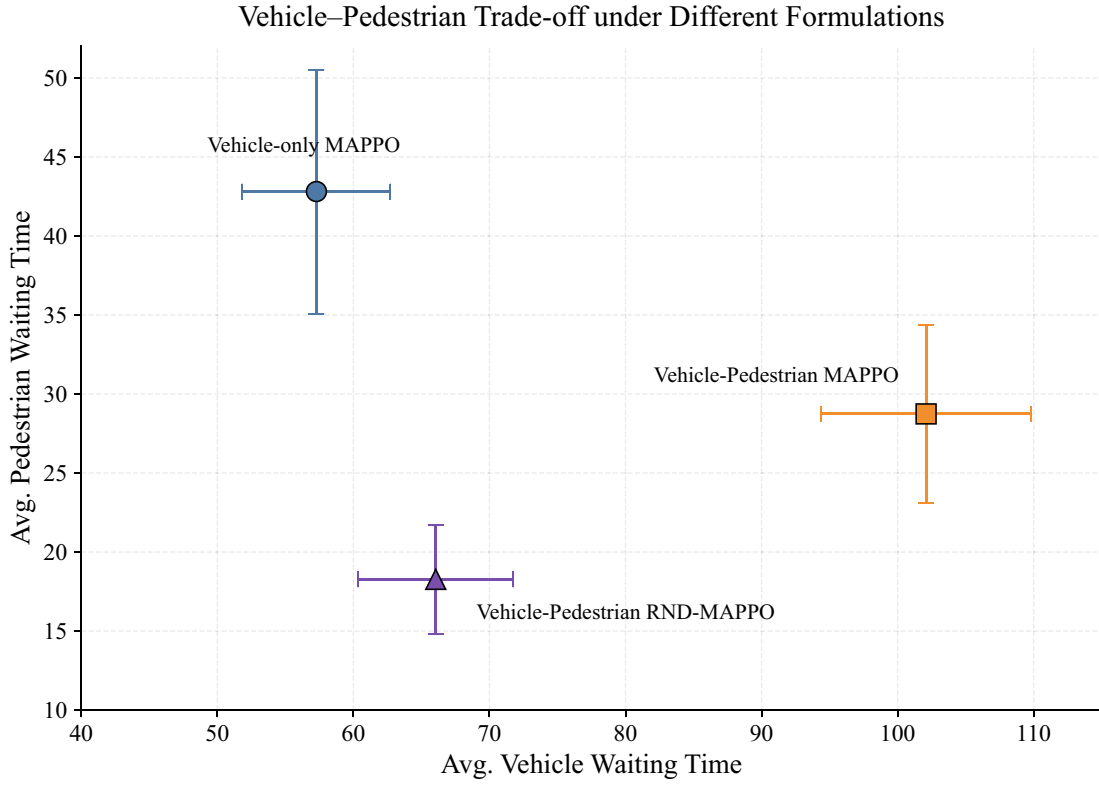


FIGURE 13 | Vehicle-pedestrian trade-off under different formulations.

TABLE 11 | Performance under different demand levels.

Demand Level	Method	Avg. queue length	Avg. vehicle waiting time	Throughput	Avg. pedestrian waiting time	Avg. waiting pedestrians
Low (0.5×)	AttendLight	37.08 ± 4.71	32.87 ± 2.82	253.78 ± 16.58	11.29 ± 2.05	5.09 ± 1.18
	MAPPO	41.51 ± 5.52	46.42 ± 3.51	221.53 ± 12.71	13.06 ± 2.55	6.44 ± 1.49
	RND-MAPPO	32.53 ± 3.82	32.22 ± 2.79	264.15 ± 17.26	8.91 ± 1.69	3.74 ± 0.94
Medium (1.0×)	AttendLight	77.86 ± 9.87	69.02 ± 5.87	532.93 ± 34.81	23.74 ± 4.31	10.69 ± 2.48
	MAPPO	91.31 ± 12.15	102.09 ± 7.73	484.01 ± 27.96	28.74 ± 5.61	14.17 ± 3.27
	RND-MAPPO	66.69 ± 7.84	66.05 ± 5.71	541.51 ± 35.39	18.27 ± 3.47	7.67 ± 1.92
High (2.0×)	AttendLight	202.44 ± 25.66	179.45 ± 15.26	1385.62 ± 90.51	61.72 ± 11.21	27.29 ± 6.45
	MAPPO	246.54 ± 32.81	275.64 ± 20.87	1306.83 ± 75.49	77.61 ± 15.15	38.26 ± 8.83
	RND-MAPPO	160.06 ± 18.82	158.52 ± 13.73	1516.23 ± 99.09	43.85 ± 8.33	18.41 ± 4.61

particularly advantageous when the control problem becomes more difficult.

A likely explanation is that the marginal value of exploration is demand-dependent. Under low demand, the network is less congested, interactions among intersections are weaker and many methods can already maintain acceptable performance. As demand increases, however, queue spillback, delayed phase effects and inter-intersection coupling become more pronounced, making the control problem harder to solve through local short-term feedback alone. In this regime, the curiosity-driven exploration signal appears to help the agents discover stronger

coordination strategies, which is why the advantage of RND-MAPPO becomes more visible under heavy traffic conditions.

5.6 | Sensitivity to the Intrinsic Reward Coefficient

The proposed method introduces an intrinsic reward coefficient λ_{int} to control the contribution of the RND-based curiosity signal during training. Since the balance between intrinsic and extrinsic rewards can affect both learning behaviour and final policy quality, we conduct a sensitivity study to examine whether the

Performance of the Compared Methods under Low, Medium, and High Demand

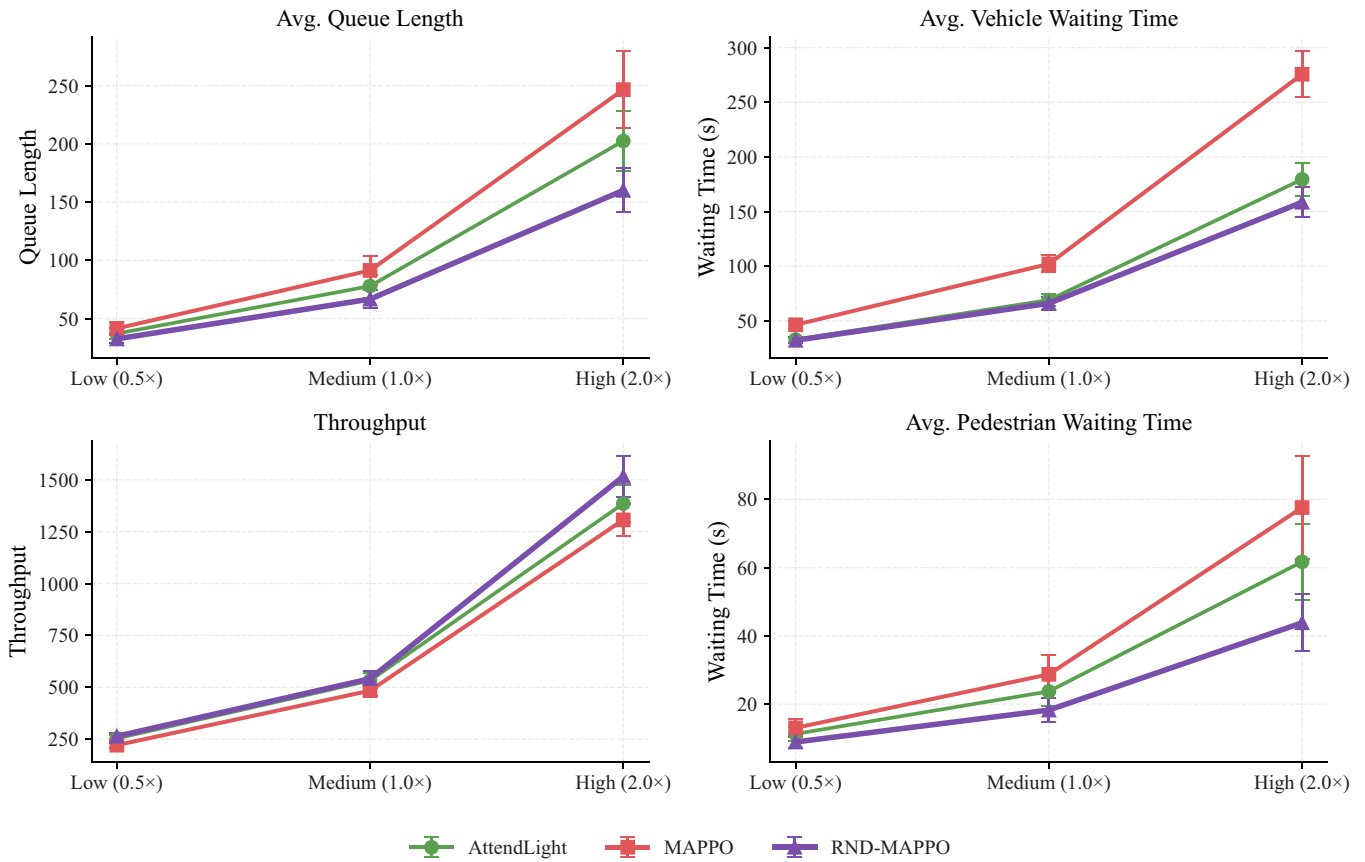


FIGURE 14 | Performance of the compared methods under low, medium and high demand.

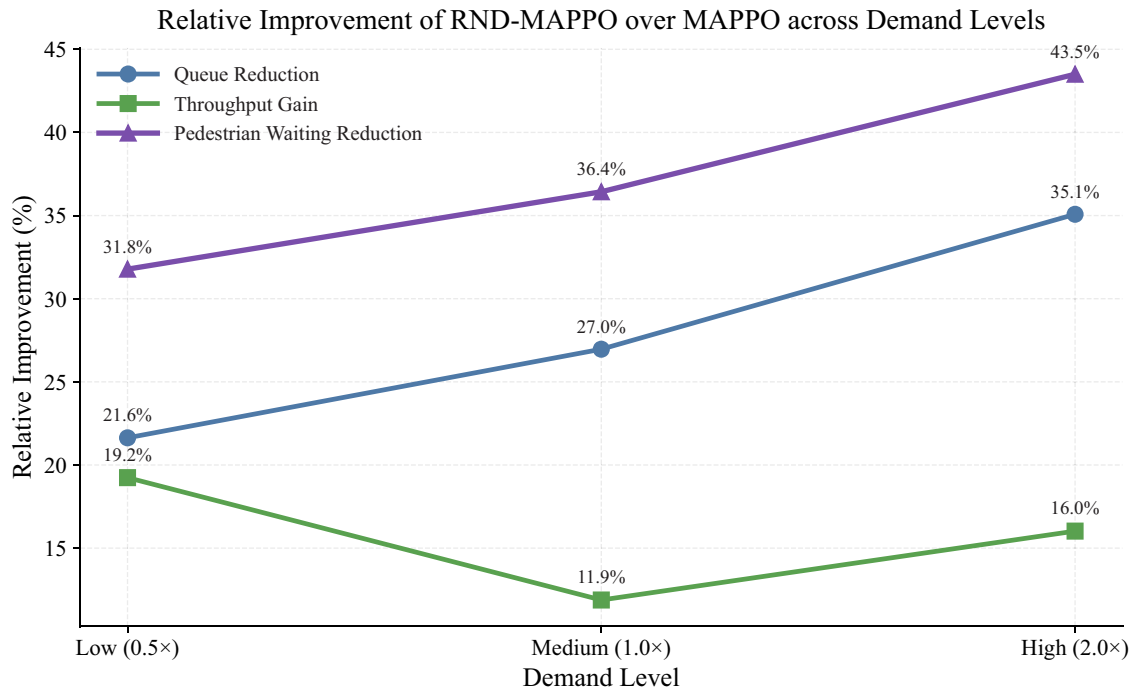
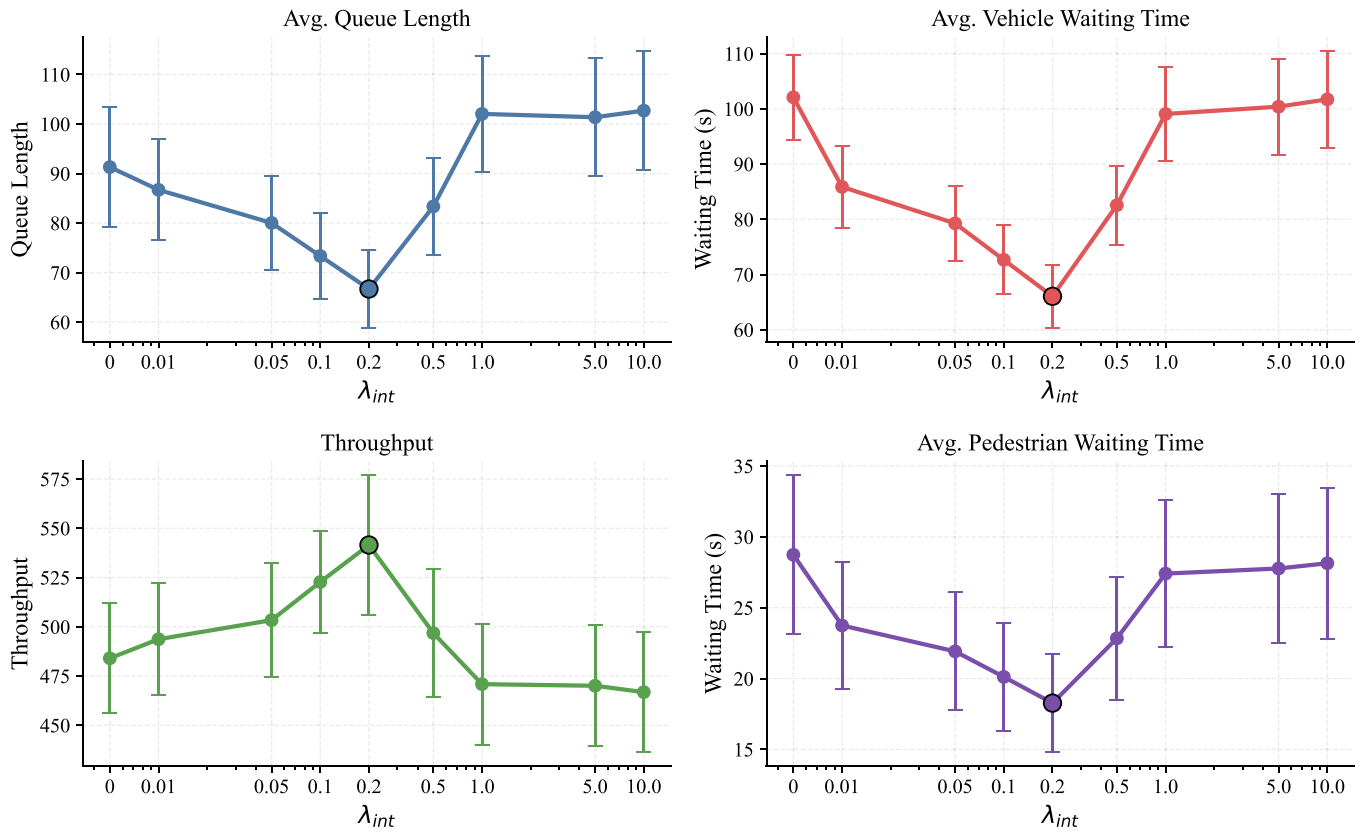


FIGURE 15 | Relative improvement of RND-MAPPO over MAPPO across demand levels.

Sensitivity of RND-MAPPO to the Intrinsic Reward Coefficient λ_{int} **FIGURE 16** | Sensitivity of RND-MAPPO to the intrinsic reward coefficient λ_{int} .**TABLE 12** | Sensitivity analysis of the intrinsic reward coefficient λ_{int} .

λ_{int}	Avg. Queue Length	Avg. Vehicle Waiting Time	Throughput	Avg. Pedestrian Waiting Time	Avg. Waiting Pedestrians
0	91.31 ± 12.15	102.09 ± 7.73	484.01 ± 27.96	28.74 ± 5.61	14.17 ± 3.27
0.01	86.71 ± 10.19	85.87 ± 7.42	493.69 ± 28.52	23.75 ± 4.51	9.97 ± 2.63
0.05	80.03 ± 9.41	79.26 ± 6.85	503.37 ± 29.08	21.92 ± 4.16	9.20 ± 2.30
0.1	73.36 ± 8.62	72.66 ± 6.28	522.73 ± 25.89	20.12 ± 3.82	8.44 ± 2.11
0.2	66.69 ± 7.84	66.05 ± 5.71	541.51 ± 35.39	18.27 ± 3.47	7.67 ± 1.92
0.5	83.36 ± 9.82	82.56 ± 7.14	496.79 ± 32.47	22.84 ± 4.34	9.59 ± 2.40
1.0	102.04 ± 11.76	99.08 ± 8.57	470.87 ± 30.77	27.41 ± 5.21	11.51 ± 2.88
5.0	101.37 ± 11.92	100.40 ± 8.68	470.06 ± 30.72	27.77 ± 5.27	11.66 ± 2.92
10.0	102.70 ± 12.07	101.72 ± 8.79	466.82 ± 30.51	28.14 ± 5.34	11.81 ± 2.96

performance of RND-MAPPO depends strongly on a narrow hyperparameter choice.

Figure 16 and Table 12 report the sensitivity of the proposed method to the intrinsic reward coefficient λ_{int} . When $\lambda_{int} = 0$, the method reduces to vanilla MAPPO and yields the weakest overall performance among the tested settings. As λ_{int} increases from zero to a moderate value, the traffic and pedestrian metrics improve consistently. In our experiments, the best overall perfor-

mance is achieved at $\lambda_{int} = 0.2$, while nearby values such as 0.1 and 0.5 still provide clear gains over the no-curiosity setting.

When λ_{int} becomes too large, however, the benefit weakens. For example, at $\lambda_{int} = 1.0, 5.0$ and 10.0 , both vehicle efficiency and pedestrian service deteriorate relative to the best setting. This suggests that excessively strong curiosity can interfere with the original traffic-control objective by overemphasising novelty at the expense of task-oriented optimisation. Overall, the results

indicate that the proposed method is effective over a reasonable range of intrinsic reward coefficients, with the best overall performance obtained at $\lambda_{\text{int}} = 0.2$ and competitive performance is still maintained at nearby values such as 0.1 and 0.5.

6 | Conclusion

This paper studied pedestrian-aware multi-intersection traffic signal control in a realistic urban road network and proposed a curiosity-enhanced cooperative reinforcement learning framework based on MAPPO and RND. The main motivation of the work was that, in network-level traffic signal control, the effect of a signal decision is often delayed, intersections are strongly coupled and local observations may provide only incomplete feedback about network-level traffic evolution. Under such conditions, relying only on extrinsic traffic rewards may limit exploration efficiency during training. To address this issue, the proposed method introduced an intrinsic-reward mechanism into MAPPO so that each intersection agent could explore less familiar traffic states more effectively while still learning under a cooperative multi-agent framework.

To better reflect realistic urban traffic conditions, pedestrian demand was explicitly incorporated into both the control formulation and the evaluation process. In this way, the traffic signal control task was modelled not only as a vehicle-efficiency problem, but as a mixed vehicle-pedestrian control problem in which the learned policy must balance multiple interacting service requirements. The resulting framework was evaluated on a Helsinki SUMO subnetwork with 16 signalised intersections and the experiments considered both overall traffic performance and learning efficiency.

The results showed that the proposed RND-MAPPO method achieved strong overall performance across the main traffic metrics, including queue length, vehicle waiting time, throughput and pedestrian-related service quality. Compared with the backbone MAPPO method, the proposed approach consistently improved both final traffic performance and learning dynamics, indicating that curiosity-driven exploration provided a meaningful benefit beyond the cooperative MARL backbone itself. The ablation studies further showed that pedestrian-aware modelling was not merely an auxiliary extension, but an important factor in learning more balanced signal control policies under mixed road-user demand. In addition, the robustness experiments under low-, medium- and high-demand settings suggested that the proposed method remained effective across different congestion levels, with especially clear advantages under heavier traffic conditions. The sensitivity analysis on the intrinsic reward coefficient also indicated that the method was effective over a reasonable range of parameter values rather than depending on a single fragile setting.

Overall, the findings of this paper suggest that improving exploration is a useful direction for pedestrian-aware multi-agent traffic signal control in realistic urban environments. Rather than modifying only the coordination architecture or the reward design, the present work shows that training efficiency itself can be an important bottleneck in network-level signal control and that curiosity-enhanced learning can help alleviate this issue.

Future work may extend this framework to larger real-world traffic networks, more diverse traffic participant types and more adaptive demand settings and may also investigate how curiosity-driven exploration can be combined with stronger structural coordination mechanisms such as graph-based communication or safety-aware signal control constraints.

Author Contributions

Ruotian Chen: project investigation and implementation, writing – original draft. **Xinhua Tang:** conceptualisation, program design, supervision, review and editing. **Laura Ruotsalainen:** conceptualisation, supervision, review and editing, funding acquisition. **Zhe Yan:** research platform setup. **Chengmi Zhou:** theory consultant, review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data available on request from the authors. The data that support the findings of this study are available from the corresponding author, [Xinhua Tang], upon reasonable request.

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